Atmosphere, Weather and Climate

Ninth Edition

Roger G. Barry and
Richard J. Chorley



Atmosphere, Weather and Climate

This book presents a comprehensive introduction to weather processes and climatic conditions around the world, their observed variability and changes, and projected future trends. Extensively revised and updated, this ninth edition retains its tried and tested structure while incorporating recent advances in the field. From clear explanations of the basic physical and chemical principles of the atmosphere, to descriptions of regional climates and their changes, the book presents a comprehensive coverage of global meteorology and climatology. In this new edition the latest scientific ideas are again expressed in a clear, non-mathematical manner.

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- Over 360 diagrams have been redrawn in full colour to improve clarity and aid understanding
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Atmosphere, Weather and Climate continues to be an indispensable source for all those studying the earth's atmosphere and world climate, whether from environmental and earth sciences, geography, ecology, agriculture, hydrology, or related disciplinary perspectives. Its pedagogic value is enhanced by several features: learning points at the opening of each chapter and discussion topics at their ending, boxes on topical subjects and on twentieth century advances in the field.

Roger G. Barry is Distinguished Professor of Geography, University of Colorado at Boulder, Director of the World Data Center for Glaciology and the National Snow and Ice Data center, and a Fellow of the Cooperative Institute for Research in Environmental Sciences.

The late **Richard J. Chorley** was Professor of Geography at the University of Cambridge.

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Preface to the ninth edition

This revised ninth edition of *Atmosphere, Weather and Climate* will prove invaluable to all those studying the earth's atmosphere and world climate, whether from environmental, atmospheric and earth sciences, geography, ecology, agriculture, hydrology or related disciplinary perspectives.

Atmosphere, Weather and Climate provides a comprehensive introduction to weather processes and climatic conditions. Since the last edition, we have added an introductory overview of the historical development of the field and its major components. Following this there is an extended treatment of atmospheric composition and energy, stressing the heat budget of the earth and the causes of the greenhouse effect. We then turn to the manifestations and circulation of atmospheric moisture, including atmospheric stability and precipitation patterns in space and time. A consideration of atmospheric and oceanic motion on small to large scales leads on to a chapter on modelling of the atmospheric circulation and climate that also presents weather forecasting on different time scales. This was revised by my colleague Dr Tom Chase of CIRES and Geography at the University of Colorado, Boulder. This is followed by a discussion of the structure of air masses, the development of frontal and non-frontal cyclones and of mesoscale convective systems in mid-latitudes. The treatment of weather and climate in temperate latitudes begins with studies of Europe and America, extending to the conditions of their subtropical and high-latitude margins and includes the Mediterranean, Australasia, North Africa, the southern westerlies, and the subarctic and polar regions. Tropical weather and climate are also described through an analysis of the climatic mechanisms of monsoon Asia, Africa, Australia and Amazonia, together with the tropical margins of Africa and Australia and the effects of ocean movement and the El Niño–Southern Oscillation and teleconnections. Small-scale climates – including urban climates – are considered from the perspective of energy budgets. The final chapter revised by Dr Mark Serreze of CIRES stresses the structure and operation of the atmosphere–earth–ocean system and the causes of its climate changes. Since the previous edition appeared in 2003, the pace of research on the climate system and attention to global climate change has accelerated. A discussion of the various modeling strategies adopted for the prediction of climate change is undertaken, in

particular relating to the IPCC 1990–2007 models. A consideration of other environmental impacts of climate change is also included.

The new information age and wide use of the World Wide Web has led to significant changes in presentation. Apart from the two revised chapters 8 and 13, new features include all figures redrawn in color. Wherever possible, the criticisms and suggestions of colleagues and reviewers have been taken into account in preparing this latest edition.

This edition benefited greatly from the ideas and work of my long-time friend and co-author Professor Richard J. Chorley, who sadly passed away on 12 May 2002. His knowledge, enthusiasm and inspiration are sorely missed.

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The redrawn color figures were prepared by Mr Paul Coles in the Geography Department at Sheffield University, UK, building on the illustrative imagination and cartographic expertise of Mr M. Young of the Department of Geography, Cambridge University. The authors owe a considerable debt of gratitude to both individuals.

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CHAPTER ONE

Introduction and history of meteorology and climatology

1

LEARNING OBJECTIVES

When you have read this chapter you will:

- be familiar with key concepts in meteorology and climatology
- know how these fields of study evolved and the contributions of leading individuals.

A THE ATMOSPHERE

The atmosphere, vital to terrestrial life, envelops the earth to a thickness of only 1 percent of the earth's radius. It had evolved to its present form and composition at least 400 million years ago by which time a considerable vegetation cover had developed on land. At its base, the atmosphere rests on the land and ocean surface, the latter which, at present, covers some 71 percent of the surface of the globe. Although air and water share somewhat similar physical properties, they differ in one important respect – air is compressible, while water is largely incompressible. In other words, in contrast to water, if one were to 'squeeze' a given sample of air, its volume would decrease. Study of the atmosphere has a long history involving observations, theory, and, since the 1960s, numerical modeling. Like most scientific fields, incremental progress has been interspesed by moments of great insight and rapid advance.

Scientific measurements only became possible with the invention of appropriate instruments; most had a long and complex evolution. A thermometer was invented by Galileo in the early 1600s, but accurate liquid-in-glass thermometers with calibrated scales were not available until the early 1700s (Fahrenheit), or 1740s (Celsius). In 1643 Torricelli invented the barometer, and demonstrated that the weight of the atmosphere at sea level would support a 10m column of water or a 760mm column of liquid mercury. Pascal used a barometer of Torricelli to show that pressure decreases with altitude, by taking one up the Puy de Dome in France. This paved the way for Boyle (1660) to demonstrate the compressibility of air by propounding his law that volume is inversely proportional to pressure. It was not until 1802 that Charles made the discovery that air volume is also directly proportional to its temperature. Combining Boyle's and Charles' laws yields the ideal gas law relating pressure, volume and temperature, one of the most important relationships in atmospheric science. By the end of the nineteenth century the four major constituents of the dry atmosphere

(nitrogen 78.08 percent, oxygen 20.98 percent, argon 0.93 percent and carbon dioxide 0.035 percent) had been identified. It had been long suspected that human activities could have the potential to alter climate. While the atmospheric 'greenhouse effect' was discovered in 1824 by Joseph Fourier, the first serious consideration of a link between climate change, the greenhouse effect and changes in atmospheric carbon dioxide also emerged in the late nineteenth century through the insights of Swedish scientist Svante Arthenius. His expectation that carbon dioxide levels and temperature would rise due to fossil fuel burning has sadly turned out to be correct.

The hair hygrograph, designed to measure relative humidity (the amount of water vapor in the atmosphere relative to how much it can hold at saturation, expressed as a percent), was invented in 1780 by de Saussure. Rainfall records exist from the late seventeenth century in England, although early measurements are described from India in the fourth century BC, Palestine about AD 100 and Korea in the 1440s. A cloud classification scheme was devised by Luke Howard in 1803, but was not fully developed and implemented in observational practice until the 1920s. Equally vital was the establishment of networks of observing stations, following a standardized set of procedures for observing the weather and its elements, and a rapid means of exchanging the data (the telegraph). These two developments went hand-in-hand in Europe and North America in the 1850s–1860s.

The greater density of water compared with that of air (a factor of about 1000 at mean sea level pressure) gives water a higher specific heat. In other words, much more heat is required to raise the temperature of a cubic meter of water by 1°C than to raise the temperature of an equal volume of air by the same amount. It is interesting to note that just the top 10–15cm of ocean waters contain as much heat as does the total atmosphere; the total heat in the ocean in turn dwarfs that of the atmosphere. As is now known, this tremendous reservoir of heat in the upper ocean and its exchanges with the atmosphere is key to understanding climate variability. Another important feature of the behavior of air and water appears during the process of evaporation or condensation. As Black showed in 1760, during evaporation, heat energy of water is translated into kinetic energy of water vapor molecules (i.e., latent heat), whereas subsequent condensation in a cloud or as fog releases kinetic energy which returns as heat energy. The amount of water which can be stored in water vapor depends on the temperature of the air. This is why the condensation of warm, moist tropical air releases large amounts of latent heat increasing the instability of tropical air masses. This may be considered as part of the process of convection in which heated air expands, decreases in density and rises, perhaps resulting in precipitation, whereas cooling air contracts, increases in density and subsides.

The combined use of the barometer and thermometer allowed the vertical structure of the atmosphere to be investigated. While it is common experience to the aviator and mountain traveler that temperature tends to decrease with height, the reverse pattern of temperature increasing with height, known as an inversion, is also quite common, and in fact dominates in certain regions and atmospheric levels. A low-level (i.e., near-surface) temperature inversion was discovered in 1856 at a height of about 1km on a mountain in Tenerife. Later investigations revealed that this so-called Trade Wind Inversion is found over the eastern subtropical oceans where subsiding dry high pressure air overlies cool, moist maritime air close to the ocean surface. Such inversions inhibit vertical (convective) air movements and, consequently, form a lid to some atmospheric activity. The Trade Wind Inversion was shown in the 1920s to differ in elevation between some 500m and 2km in different parts of the Atlantic Ocean in the belt 30°N to 30°S. Around 1900 a more important continuous and widespread temperature inversion was revealed by balloon flights to exist at about 10km at the equator and 8km at high latitudes. This inversion

level (the tropopause) was recognized to mark the top of the so-called troposphere within which most weather systems form and decay. By 1930 balloons equipped with an array of instruments to measure pressure, temperature and humidity, and report them back to earth by radio (radiosonde), were routinely investigating the atmosphere. Observations from both kites and balloons also revealed that strong inversions extending up to about 1000m are a near ubiquitous feature of the Arctic in winter.

B SOLAR ENERGY

Differential solar heating of low and high latitudes is the mechanism which drives the earth's large-scale atmospheric and oceanic circulations. Most of the energy from the sun entering the atmosphere as short-wave radiation (or insolation) reaches the earth's surface. Some is reflected back to space. The remainder is absorbed by the surface which then warms the atmosphere above it. The atmosphere and surface together radiate long-wave (thermal) radiation back to space. Although the land and ocean parts of the surface absorb different amounts of solar radiation and have different thermal characteristics, the differential solar heating between low and high latitudes dominates, fostering an equator-to-pole gradient in atmospheric and upper ocean temperatures.

Although increased solar heating of the tropical regions compared with the higher latitudes had long been apparent, it was not until 1830 that Schmidt made a key calculation, namely heat gains and losses for each latitude by incoming solar radiation and by outgoing longwave radiation from the earth. This showed that equatorward of about latitudes 35° there is an excess of incoming solar over outgoing longwave energy, while poleward of those latitudes the longwave loss exceeds solar input. If, at each latitude, the longwave loss to space equaled the solar radiation input (termed radiative equilibrium), this pattern would not be seen. That it exists is direct evidence that there must be an overall tranasfer of energy from lower to higher latitudes via the atmospheric and oceanic circulations. Put differently, while the differential solar heating gives rise to the equator-to-pole temperature gradient, the poleward energy transports work to reduce this gradient. Later and more refined calculations showed that the poleward flow (or flux) of atmospheric energy reaches a maximum around latitudes 30° and 40°, with the maximum ocean transport occurring at lower latitudes. The total poleward transport in both hemispheres is in turn dominated by the atmosphere. The amount of solar energy being received and re-radiated from the earth's surface can be computed theoretically by mathematicians and astronomers. Following Schmidt, many such calculations were made, notably by Meech (1857), Wiener (1877) and Angot (1883) who calculated the amount of extraterrestrial insolation received at the outer limits of the atmosphere at all latitudes. Theoretical calculations of insolation in the past by Milankovitch (1920, 1930), and Simpson's (1928–1929) calculated values of the insolation balance over the earth's surface, were important contributions to understanding astronomic controls of climate. Nevertheless, the solar radiation received by the earth was only accurately determined by satellites in the 1990s.

C GLOBAL CIRCULATION

While differential solar heating of the surface and the atmospheric temperature gradient that it generates fosters the large-scale transport of energy from equatorial to polar regions, what are the mechanisms by which this atmospheric transport is accomplished? While we now know that the transport is accomplished by the Hadley circulation in lower latitudes and in higher latitudes through disturbances in the basic westerly (west to east) flow in the form of transient cyclones and anticyclones, it is fascinating to briefly outline how our modern view of the global circulation emerged.

The first attempt to explain the global atmospheric circulation was based on a simple convectional concept. In 1686 Halley associated the easterly Trade Winds with low-level convergence on the equatorial belt of greatest heating (i.e., the thermal equator). These flows are compensated at high levels by return flows aloft. Poleward of these convectional regions, the air cools and subsides to feed the northeasterly and southeasterly Trade Winds at the surface. This simple mechanism, however, presented two significant problems: what mechanism produced the observed high pressure in the subtropics and what was responsible for the belts of dominantly westerly winds poleward of this high pressure zone? It is interesting to note that it was not until 1883 that Teisserenc de Bort produced the first global mean sea-level map showing the main zones of high and low pressure. The climatic significance of Halley's work rests also in his thermal convectional theory for the origin of the Asiatic monsoon which was based on the differential thermal behavior of land and sea; i.e., the land reflects more and stores less of the incoming solar radiation and therefore heats and cools faster. This heating causes continental surface pressures to be generally lower than oceanic ones in summer and higher in winter, causing seasonal wind reversals. The role of seasonal movements of the thermal equator in monsoon systems was only recognized much later. Some of the difficulties faced by Halley's simplistic large-scale circulation theory began to be addressed by Hadley in 1735, who was particularly concerned with the deflection of winds on a rotating globe, to the right (left) in the Northern (Southern) Hemisphere. Like Halley, he advocated a thermal circulatory mechanism, but was perplexed by the existence of the westerlies. Following the mathematical analysis of moving bodies on a rotating earth by Coriolis (1831), Ferrel (1856) developed a three-cell model of hemispherical atmospheric circulation by suggesting a mechanism for the production of high pressure in the subtropics (i.e., 35°N and S latitude). The tendency for cold upper air to subside in the subtropics, together with the latitudinal increase in the deflective force (the Coriolis force, the product of wind speed and the the Coriolis parameter which increases with latitude) applied by terrestrial rotation to upper air moving poleward above the Trade Wind Belt, would cause a buildup of air (and therefore of pressure) in the subtropics. Equatorward of these subtropical highs the thermally direct Hadley cells dominate the Trade Wind Belt but poleward of them air tends to flow towards higher latitudes at the surface. This airflow, increasingly deflected with latitude, constitutes the westerly winds in both hemispheres. In the Northern Hemisphere, the highly variable northern margin of the westerlies is situated where the westerlies are undercut by polar air moving equatorward. This margin was compared with a battlefield front by Bergeron who, in 1922, termed it the Polar Front. Thus, Ferrel's three cells consisted of two thermally direct Hadley cells (where warm air rises and cool air sinks), separated by a weak, indirect Ferrel cell in mid-latitudes. The relation between pressure distribution and wind speed and direction was demonstrated by Buys-Ballot in 1860.

D CLIMATOLOGY

During the nineteenth century it became possible to assemble a large body of global climatic data and to use it to make useful regional generalizations. In 1817 Alexander von Humboldt produced his valuable treatise on global temperatures containing a map of mean annual isotherms (lines of equal temperature) for the Northern Hemisphere but it was not until 1848 that Dove published the first world maps of monthly mean temperature. An early world map of precipitation was produced by Berghaus in 1845; in 1882 Loomis produced the first world map of precipitation employing mean annual isohyets (lines of equal precipitation); and in 1886 de Bort published the first world maps of annual and monthly cloudiness.

These generalizations allowed, in the later decades of the century, attempts to classify climates regionally. In the 1870s Wladimir Koeppen, a St Petersburg-trained biologist, began producing maps of climate based on plant geography, as did de Candolle (1875) and Drude (1887). In 1883 Hann's massive, three-volume *Handbook of Climatology* appeared, which remained a standard until 1930–1940 when the five-volume work of the same title by Koeppen and Geiger replaced it. At the end of World War I Koeppen (1918) produced the first detailed classification of world climates based on terrestrial vegetation cover. This was followed by Thornthwaite's (1931–1933) classification of climates employing evaporation and precipitation amounts, which he made more widely applicable in 1948 by the use of the theoretical concept of potential evapo-transpiration. The Inter-War period was particularly notable for the appearance of a number of climatic ideas which were not brought to fruition until the 1950s. These included the use of frequencies of various weather types (Federov 1921), the concepts of variability of temperature and rainfall (Gorczynski 1942 and 1945) and microclimatology, the study of the fine climate structure near the surface (Geiger 1927).

Despite the problems of obtaining detailed measurements over the large ocean areas, the later nineteenth century saw much climatic research which was concerned with pressure and wind distributions. In 1868 Buchan produced the first world maps of monthly mean pressure; eight years later Coffin composed the first world wind charts for land and sea areas, and in 1883 L. Teisserenc de Bort produced the first mean global pressure maps showing the cyclonic and anticyclonic 'centers of action' on which the general circulation is based. In 1887 de Bort began producing maps of upper-air pressure distributions and in 1889 his world map of January mean pressures in the lowest 4km of the atmosphere was particularly effective in depicting the great belt of the westerlies between 30° and 50° north latitudes.

E MID-LATITUDE DISTURBANCES

Theoretical ideas about the atmosphere and its weather systems evolved in part through the needs of nineteenth-century mariners for information about winds and storms, especially predictions of future behavior. At low levels in the westerly belt (approximately 40° to 70° latitude) there is a complex pattern of moving high and low pressure systems, while between 6000m and 20,000m there is a coherent westerly airflow. Dove (1827 and 1828) and Fitz Roy (1863) supported the 'opposing current' theory of cyclone (i.e., depression) formation, where the energy for the systems was produced by converging airflow. Espy (1841) set out more clearly a convection theory of energy production in cyclones with the release of latent heat (condensation of water vapor) as the main source. In 1861 Jinman held that storms develop where opposing air currents form lines of confluence (later termed 'fronts'). Ley (1878) gave a three-dimensial picture of a low pressure system with a cold air wedge behind a sharp temperature discontinuity cutting into warmer air, and Abercromby (1883) described storm systems in terms of a pattern of closed isobars (lines of equal pressure) with typical associated weather types. By this time, although the energetics were far from clear, a picture, correct in is basics, was emerging of mid-latitude storms being generated by the mixing of warm tropical and cool polar air as a fundamental result of the latitudinal temperature gradients created by the patterns of incoming solar radiation and of outgoing terrestrial radiation. Towards the end of the nineteenth century two important European research groups were dealing with storm formation: the Vienna Group under Margules, including Exner and Schmidt; and the Swedish Group led by Vilhelm Bjerknes. The former workers were concerned with the origins of cyclone kinetic energy (energy of motion) which was thought to be due to differences in the potential energy of opposing air masses of different temperature. Potential energy is energy

associated with the height of air parcels above the surface. Gradients in potential energy on a pressure surface provide conditions to convert potential to kinetic energy. This was set forth in the work of Margules (1901) who showed that the potential energy of a typical depression is less than 10 percent of the kinetic energy of its constituent winds. In Stockholm V. Bjerknes' group concentrated on frontal development (Bjerknes, 1897 and 1902) but its researches were particularly important during the period 1917–1929 after J. Bjerknes moved to Bergen and worked with Bergeron. In 1918 the warm front was identified, the occlusion process was described in 1919, and the full Polar Front Theory of cyclone development was presented in 1922 (J. Bjerknes and Solberg). After about 1930, meteorological research concentrated increasingly on the importance of mid- and upper-tropospheric influences for global weather phenomena. This was led by Sir Napier Shaw in Britain and by Rossby, with Namias and others, in the USA. The airflow in the 3–10km high layer of the polar vortex of the Northern Hemisphere westerlies was shown to form large-scale horizontal (Rossby) waves due to latitudunal gradients in the Coriolis parameter, the influence of which was simulated by rotating 'dish pan' experiments in the 1940s and 1950s. The number and amplitude of these waves appear to depend on the hemispheric energy gradient, or 'index'. At times of high index, especially in winter, there may be as few as three Rossby waves of small amplitude giving a strong zonal (i.e., west to east) flow. A weaker hemispheric energy gradient (i.e., low index) is characterized by four to six Rossby waves of larger amplitude. As with most broad, fluid-like flows in nature, the upper westerlies were shown by observations in the 1920s and 1930s, and particularly by aircraft observations in World War II, to possess narrow high-velocity threads, termed 'jet streams' by Seilkopf in 1939. The higher and more important jet streams approximately lie along the Rossby waves. The most important jet stream, located at 10km, clearly affects surface weather by guiding the low pressure systems which tend to form beneath it. In addition, air subsiding beneath the jet streams strengthens the subtropical high pressure cells.

F THE POLAR REGIONS

The earliest view of the Arctic's atmospheric circulation can be traced to the late nineteenth-century work of von Helmholtz, who argued that the region was dominated by a more or less permanent surface high pressure cell, a view developed in the earlier part of the twentieth century by Hobbs in his 'glacial anticyclone' theory. In 1945, Hobbs elaborated further on this basic idea, advocating the existence of a persistent anticyclone over the Greenland ice sheet, having strong impacts on middle latitudes. Given the general lack of data until the 1940s and 1950s, such a misconception is not surprising. Sea level pressure analyses in the US Historical Weather Map Series produced during World War II contained strong positive biases prior to the 1930s away from the North Atlantic sector. Part of the problem, as noted by Jones, was that these maps were prepared by relatively untrained analysts, who tended to extrapolate into the data-poor Arctic with the prevailing mindset of an Arctic high pressure cell. Even by the early 1950s, some studies erroneously depicted traveling cyclones as largely restricted to the periphery of the Arctic Ocean. The emergence in North America of more modern views of the Arctic circulation in the late 1950s and 1960s, fostered by the growing database of upper-air data and surface observations, appeared in the work of research groups at McGill University led by F. K. Hare and the University of Washington led by R. J. Reed. R. G. Barry participated in the work at McGill, and made many contributions. Interestingly, in the Soviet Union, a relatively modern view of the summer circulation had already been formulated in 1945 by B. L. Dzerdzeevskii.

Knowledge of Antarctica lagged behind that of the Arctic. The remoteness and extremely harsh

conditions of this continent were barriers to progress. Furthermore, while the Arctic was a key strategic region during the Cold War, leading to extensive research and the rapid establishment of observing networks, the Antarctic did not benefit from Cold War activity to the same degree. Some features had long been recognized, such as the existence of a trough of low pressure surrounding the continent, and persistent strong katabatic (downslope) winds. Much progress was made following extensive observations made during the International Geophysical Year (IGY) of 1957–1958, which was modeled on the International Polar Years of 1882–1883 and 1932–1933 (Box 1.1). A preliminary survey of the Southern Hemisphere westerlies, based in part on upper-air observations during the IGY, was published by H.H. Lamb in 1959. Even today, direct observations are much more sparse over the Antarctic than over the Arctic. Weather forecasts in this region rely especially strongly on data collected by earth-orbiting satellites.

G TROPICAL WEATHER

The success in modeling the life cycle of the mid-latitude frontal depression, and its value as a forecasting tool, naturally led to attempts in the immediate pre-World War II period to apply it to the atmospheric conditions which dominate the tropics (i.e., 30°N–30°S), comprising half the surface area of the globe. This attempt was largely doomed to failure, as observations made during the air war in the Pacific soon demonstrated. This failure was due to the lack of frontal temperature discontinuities between air masses and the absence of a strong Coriolis effect and thus of Rossby-like waves. Tropical airmass discontinuities are based on moisture differences. Tropical weather results mainly from strong convectional features such as heat lows, tropical cyclones (hurricanes and typhoons) and the Inter-Tropical Convergence Zone (ITCZ), the axis of which represents the dividing line between the southeast and northeast Trade Winds of the Northern and Southern Hemisphere. The huge instability of tropical airmasses means that even mild convergence in the Trade Winds gives rise to atmospheric waves travelling westward with characteristic weather patterns.

Above the Pacific and Atlantic Oceans the ITCZ is quasi-stationary with a latitudinal displacement annually of 5° or less, but elsewhere it varies between latitudes 17°S and 8°N in January and between 2°N and 27°N in July – i.e., during the southern and northern summer monsoon seasons, respectively. The seasonal movement of the ITCZ and the existence of other convective influences make the south and east Asian monsoon the most significant seasonal global weather phenomenon.

Investigations of weather conditions over the broad expanses of the tropical oceans were assisted by satellite observations after about 1960. Observations of waves in the tropical easterlies began in the Caribbean during the mid-1940s, but the structure of meso-scale cloud clusters and associated storms was only recognized in the 1970s. Satellite observations also proved very valuable in detecting the generation of hurricanes over the great expanses of the tropical oceans.

In the late 1940s and subsequently, important work was conducted on the relations between the south Asian monsoon mechanism concerning the westerly subtropical jet stream, the Himalayan mountain barrier and the displacement of the ITCZ. The very significant failure of the Indian summer monsoon in 1877 had led Blanford (1860) in India, Todd (1888) in Australia, and others, to seek correlations between Indian monsoon rainfall and other climatic phenomena such as the amount of Himalayan snowfall (which influences large-scale differential heating between the land and ocean) and the strength of the southern Indian Ocean high pressure center. Such correlations were studied intensively by Sir Gilbert Walker and his co-workers in India between about 1909 and the late 1930s. In 1924 a major advance was made when Walker identified the 'Southern Oscillation' – an east–west seesaw of atmospheric pressure and resulting rainfall (i.e., negative

correlation) between Indonesia and the eastern Pacific. Other north–south climatic oscillations were identified in the North Atlantic (Azores vs. Iceland, known as the North Atlantic Oscillation) and the North Pacific (Alaska vs. Hawaii). In the phase of the Southern Oscillation when there is high pressure over the eastern Pacific, westward-flowing central Pacific surface waters, with a consequent upwelling of cold water, plankton-rich, off the coast of South America, are associated with ascending air giving heavy summer rains over Indonesia. Periodically, weakening and breakup of the eastern Pacific high pressure cell leads to important consequences. The chief among these are subsiding air and drought over India and Indonesia and the removal of the mechanism of the cold coastal upwelling off the South American coast with the consequent failure of the fisheries there. The presence of warm coastal water is termed 'El Niño'. Although the central role played by lower latitude high pressure systems over the global circulations of atmosphere and oceans is well recognized, the cause of the east Pacific pressure change which gives rise to El Niño is not yet fully understood. There was a waning of interest in the Southern Oscillation and associated phenomena during the 1940s to mid-1960s, but the work of Berlage (1957), the increase in the number of Indian droughts during the period 1965–1990, and especially the strong El Niño which caused immense economic hardship in 1972, led to a revival of interest and research. One feature of this research has been the thorough study of the 'teleconnections' (correlations between climatic conditions in widely separated regions of the earth) pointed out by Sir Gilbert Walker.

H PALEOCLIMATES

Before the middle of the twentieth century thirty years of records was generally regarded as sufficient in order to define a given climate. By the 1960s the idea of a static climate was recognized as being untenable. New approaches to paleoclimatology, the study of past climates, were developed in the 1960s to 1970s. The astronomical theory to explain the great ice ages of the Pleistocene proposed by Croll (1867), and developed mathematically by Milankovitch (1920), seemed to conflict with evidence for dated climate changes. However, in 1976, Hays, Imbrie and Shackleton recalculated Milankovitch's chronology using powerful new statistical techniques and showed that it correlated well with past temperature records, especially for ocean paleotemperatures derived from isotopic ($^{18}0/^{16}0$) ratios in marine organisms recorded in ocean cores. The idea behind Milankovitch forcings is that periodic changes in the eccentricity of the earth's orbit, the tilt of the earth's axis and the timing of the equinoxes cause variations in the amount of solar radiation received at different times of the year over different parts of the surface. As is now widely accepted, the major ice ages as well as intervening interglacials over about the past two million years reflect influences of these Milankovitch cycles and attendant climate feeedbacks that amplify change. Paleoclimate information from ocean cores and terrestrial sources is complemented by ice cores obtained from the Greenland and Antarctic ice sheets, ice caps in Canada and elsewhere. As well as documenting climate links with Milankovitch cycles, these records provide evidence of rapid, large-scale shifts in climate. The longest ice core record currently available from Dome C in the eastern Antarctic spans 800,000 years and shows that interglacials before 450,000 years ago were weaker (less warm) than subsequently. Reconstructed temperature records from ice cores are obtained from oxygen isotope ratios ($\partial^{18}O$). Samples of past atmospheres trapped as bubbles in ice cores also document close links between climate and atmospheric carbon dioxide concentrations, and show convincingly that present-day concentrations of this greenhouse gas are higher than at any time during at least the past 800,000 years.

Other paleoclinatic information is obtained from annual tree rings that reflect growing season

temperature and moisture, lake and peat bog sediments that contain pollen records of regional vegetation, reconstructed temperature records from oxygen isotope ratios in cave stalagmite and annual growth rings in ocean corals.

Major advances have been made in paleoclimate reconstruction through the use of general circulation models with past boundary conditions (paleogeography, paleovegetation) and changed earth orbital characteristics.

I THE GLOBAL CLIMATE SYSTEM

Undoubtedly the most important outcome of work in the second half of the twentieth century was the recognition of the existence of the global climate system (see Box 1.1). The climate system involves not just the atmosphere elements, but the five major subsystems: the atmosphere (the most unstable and rapidly changing); the ocean (very sluggish in terms of its thermal inertia and therefore important in regulating atmospheric variations); the snow and ice cover (the cryosphere); and the land surface with its vegetation cover (the lithosphere and biosphere). Physical, chemical and biological processes take place in and among these complex subsystems. The most important interaction takes place between the highly dynamic atmosphere, through which solar energy is input into the system, and the oceans which store and transport large amounts of energy (especially thermal), thereby acting as a regulator to more rapid atmospheric changes. A further complication is provided by the living matter of the biosphere, which influences the incoming radiation and outgoing re-radiation and affects the atmospheric composition via greenhouse gases. In the oceans, marine biota play a major role in the dissolution and storage of CO_2. All subsystems are linked by fluxes of mass, heat and momentum into a very complex whole. The coupled climate system always has and always will be characterized by variability on numerous time and space scales. However, the introduction of humans into the system has added a new dimension. Indeed, by the dawn of the twenty-first century, overwhelming evidence had amassed of a discernible and growing human impact upon global climate.

The driving mechanism of global climate change is referred to as 'radiative forcing'. In an equilibrium climate state, globally averaged solar energy absorbed by the earth system is balanced by the globally averaged longwave radiation emitted to space. In other words, there is radiative equilibrium at the top of the atmosphere. An imbalance, or radiative forcing, is defined as positive when less energy is emitted than absorbed and negative for the reverse. In response to radiative forcing, the system tries to come back to a new equilibrium, attended by, respectively warming or cooling at the surface. Radiation imbalances arise from both natural processes (e.g., astronomical effects on incoming shortwave solar radiation, changes in total solar output and volcanic eruptions, the latter of which load the atmosphere with aerosols, tiny particles suspended in the air) and human influences (e.g., changes in greenhouse gas and aerosol concentrations owing to fossil fuel burning and a suite of other activities, such as deforestation and agriculture). Direct solar radiation measurements have been made via satellites since about 1980, but the correlation between small changes in solar radiation and in the thermal economy of the global climate system is still somewhat unclear. However, observed human-induced increases in the greenhouse gas content of the atmosphere (0.1 percent of which is composed of the trace gases carbon dioxide, methane, nitrous oxide and ozone) appear to have been very significant in increasing the proportion of terrestrial longwave radiation trapped by the atmosphere (a positive radiative forcing), thereby raising surface air temperature over the past 100 years.

Adjustments to a radiative forcing take place in a matter of months in the surface and tropospheric subsystems but are slower (centuries or longer) in the ocean. In turn, the amount of surface warming

1.1 Global Atmospheric Research Programme (GARP) and the World Climate Research Programme (WCRP)

The idea of studying global climate through coordinated intensive programs of observations emerged through the World Meteorological Organization

(WMO:http://www.wmo.ch/) and the International Council on Science (ICSU: http://www.icsu.org) in the 1970s. Three 'streams' of activity were planned: a physical basis for long-range weather forecasting; interannual climate variability; and long-term climatic trends and climate sensitivity. Global meteorological observations became a major concern and this led to a series of observational programs. The earliest was the Global Atmospheric Research Programme (GARP). This had a number of related but semi-independent components. One of the earliest was the GARP Atlantic Tropical Experiment (GATE) in the eastern North Atlantic, off West Africa, in 1974–1975. The objectives were to examine the structure of the Trade Wind inversion and to identify the conditions associated with the development of tropical disturbances. There was a series of monsoon experiments in West Africa and the Indian Ocean in the late 1970s to early 1980s and also an Alpine Experiment. The First GARP Global Experiment (FGGE), in November 1978 to March 1979, assembled global weather observations. Coupled with these observational programs, there was also a coordinated effort to improve numerical modeling of global climate processes.

The World Climate Research Programme (WCRP: http://www.wmo.ch/web/wcrp/prgs.htm), established in 1980, is sponsored by the WMO, ICSU and the International Ocean Commission (IOC). The first major global effort was the World Ocean Circulation Experiment (WOCE) which provided detailed understanding of ocean currents and the global thermohaline circulation. This was followed in the 1980s by the Tropical Ocean Global Atmosphere (TOGA).

Current major WCRP projects are Climate Variability and Predictability (CLIVAR: http://www.clivar.org/), the Global Energy and Water Cycle Experiment (GEWEX), Stratospheric Processes and their Role in Climate (SPARC), and Climate and Cryosphere (CliC; http://clic.npolar.no/) . Under GEWEX are the International Satellite Cloud Climatology Project (ISCCP) and the International Land Surface Climatology Project (ISLSCP) which provide valuable data sets for analysis and model validation. CliC, which addresses all major components of the earth's cryosphere (glaciers, ice caps and ice sheets, sea ice, snow cover, seasonally frozen ground and permafrost, the latter representing perennially frozen ground) developed from the earlier Arctic Climate System (ACSYS) effort. The WCRP has also been activity involved in the planning and implementation of the third International Polar Year (IPY), a large international scientific program focused on the Arctic and Antarctic from March 2007 to March 2009.

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to a given radiative forcing (termed climate sensitivity) depends critically on feedbacks that amplify or dampen the climate response to the forcing. In the case of greenhouse gases, the issue is further complicated in that the radiative forcing is itself changing. Major feedbacks involve the role of snow and ice reflecting incoming solar radiation and atmospheric water vapor absorbing terrestrial re-radiation, and are positive in character. For example: the earth warms; atmospheric water vapor increases; this, in turn, increases the greenhouse effect; the result being that the earth warms further. Similar warming occurs as higher temperatures reduce snow and ice cover allowing the land or ocean to absorb more radiation. Clouds play a more complex and still incompletely understood role by reflecting solar (shortwave radiation) but also by trapping terrestrial outgoing radiation. Negative feedback, when the effect of change is damped down, is a much less important feature of the operation of the climate system, which partly explains the tendency to recent global warming. The impact of aerosols is one of the biggest areas of uncertainty. While the cooling effect of aerosols through scattering solar radiation back to space is well known, and in part masks the warming effect of greenhouse gases, some aerosols, such as soot, absorb solar radiation. Aerosols also affect the number and density of cloud droplets, changing the optical properties of clouds.

An important factor in weather and climate processes is unpredictabiliy. Weather systems display sensitivity to their initial conditions, meaning that a very small change in the initial state of a weather system can have a large, disproportionate effect on the whole system. This was first recognized by E. Lorenz (1963), who pointed out that a butterfly flapping its wings in Beijing could affect the weather thousands of miles away some days later. This sensitivity is now called the 'butterfly effect'. It is addressed in numerical model experiments by running many simulations with minute variations in the initial conditions and then examining the results of an ensemble of projections.

The Intergovernmental Panel on Climate Change (IPCC), jointly established in 1988 by the WMO and the United Nations Environmental Programme (UNEP), has served as a focal point for climate change research, and released its Fourth Assessment Report in 2007. One of the most important tools of the IPCC is numerical models of the climate system. Since the initial development of atmospheric general circulation models (GCMs) in the 1960s, the current models have become very sophisticated, and are essential for untangling the complexities of radiative forcing, feedbacks and climate response. They now incorporate coupled ocean, land and biosphere submodels. The emerging picture that these models paint is of a much wamer and different world by the end of this century, posing challenges for society that include, but are not limited to, higher sea levels and shifts in agricultural zones. Major uncertainties nevertheless remain, particularly of climate change on regional scales.

The first edition of *Atmosphere, Weather and Climate* appeared in 1968 before many of the advances described in later editions were even conceived. However, our continuous aim in writing it is to provide a largely non-technical account of how the atmosphere works, thereby helping the understanding of both weather phenomena and of global climates. As noted in the eighth edition, greater explanation inevitably results in an increase in the range of phenomena requiring explanation. As a result, this book continues to thicken with time.

- How have technological advances contributed to the evolution of meteorology and climatology?
- Consider the relative contributions of observation, theory, and modeling to our knowledge of atmospheric processes.

REFERENCES AND FURTHER READING

Books

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1 Introduction and history of meteorology and climatology

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2 Atmospheric composition, mass and structure

READING

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The atmosphere is a mixture of gases with constant proportions of up to 80km or more. The

exceptions are ozone, which is concentrated in the lower stratosphere, and water vapor in the lower

troposphere. The principal greenhouse gas is water vapor. Carbon dioxide, methane and other trace

gases have increased significantly since the Industrial Revolution, especially in the twentieth

century, due to the combustion of fossil fuels, industrial processes, and other anthropogenic

effects, but larger natural fluctuations occurred during the geologic past. Reactive gases include nitrogen and sulfur and chlorine species. These play important roles in

acid precipitation and ozone destruction. Acid precipitation (by wet or dry deposition) results from

the reaction of cloud droplets with emissions of SO 2 and NOx. There are large geographical

variations in acid deposition. The processes leading to the destruction of stratospheric ozone are

complex, but the roles of nitrogen oxides and chlorine radicals are highly important in causing polar

ozone holes. Aerosols in the atmosphere originate from natural and anthropogenic sources and

they play an important but complex role in climate. Air is highly compressible, so that half of its mass occurs in the lowest 5km, and pressure

decreases logarithmically with height from an average sea-level value of 1013mb. The vertical

structure of the atmosphere comprises three relatively warm layers – the lower troposphere, the

stratopause and the upper thermosphere – separated by a cold layer above the tropopause (in the

lower stratosphere), and the mesopause. The temperature profile is determined by atmospheric

absorption of solar radiation, and the decrease of density with height.

• What properties distinguish the different layers of the atmosphere?

• What differences would exist in a dry atmosphere compared with the real atmosphere?

• What role is played by water vapor, ozone, carbon dioxide, methane and CFCs in the radiation balance of the atmosphere?

• Given the strong pressure gradient upward from the surface, why is there no large-scale upward flow of air?

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3 Solar radiation and the global energy budget

• Explain the respective roles of the earth’s orbit around the sun and the tilt of the axis of rotation for global climate.

• Explain the differences between the transmission of solar and terrestrial radiation by the atmosphere.

• What is the relative importance of incoming solar radiation, turbulent energy exchanges and other factors in determining local daytime temperatures?

• Consider the role of clouds in global climate from a radiative perspective.

• What effects do ocean currents have on regional climates? Consider the mechanisms involved for both warm and cold currents.

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4 Atmospheric moisture budget

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Brutsaert, W. (1982) Evaporation into the Atmosphere: Theory, History and Applications. Kluwer, Dordrecht, 279pp. [Thorough survey of evaporation processes and applications]

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• Trace the possible paths of a water molecule through the hydrological cycle and consider the measurements that need to be made to determine the quantities of water involved in the various transformations.

• What processes lead to phase changes of water in the atmosphere and what are some of their consequences?

• What is the significance of clouds in the global water balance?

• Compare the moisture balance of an air column and that of a small drainage basin.

• What are the various statistics used to characterize rainfall events and for what different applications are they important?

• Consider how an annual water budget diagram might differ between a wet year and a dry year at the same location. Soils, Harpenden, 124pp. [A survey of the literature on the effects of vegetation on the hydrological cycle through interception, evapotranspiration, infiltration and runoff, and of related catchment experiments around the world] Petterssen, S. (1969) Introduction to Meteorology, 3rd edn, McGraw-Hill, New York, 416pp. Rudolf, B. and Rubel, F. (2005) Global precipitation, in Hantel, M. (ed.) Observed Global Climate. Springer, Berlin, pp. 11.1–11.43 [An up-to-date overview] Sellers, W. D. (1965) Physical Climatology, University of Chicago Press, Chicago, IL, 272 pp. Strangeways, I. C. (2003) Measuring the Natural Environment , 2nd edn. Cambridge University Press, Cambridge, 548pp. [A complete account of all kinds of instrumentation] Sumner, G. (1988) Precipitation. Process and Analysis. J. Wiley and Sons, Chichester, 455 pp. [Comprehensive discussion of cloud and precipitation formation, precipitation systems, surface measurements and their analysis in time and space] World Meteorological Organization (1972) Distribution of Precipitation in Mountainous Areas (2 vols). WMO No. 326, Geneva, 228 and 587pp. [Conference proceedings with many valuable papers] Journal articles Acreman, M. (1989) Extreme rainfall in Calderdale, 19 May 1989. Weather 44, 438–46. Agnew, C. T. and Chappell, A. (2000). Desiccation in the Sahel, in McLaren, S. J. and Kniveton, D. R. (eds), Linking Climate Change to Land Surface Changes , Kluwer, Dordrecht, 27–48. Armstrong, C. F. and Stidd, C. K. (1967). A moisture balance profile on the Sierra Nevada. J. Hydrol. 5, 258–68. Atlas, D., Chou, S-H. and Byerly, W. P. (1983) The influence of coastal shape on winter mesoscale air-sea interactions. Monthly Weather Review 111, 245–52. Ayoade, J. A. (1976) A preliminary study of the magnitude, frequency and distribution of intense rainfall in Nigeria. Hydro. Sci. Bull. 21(3), 419–29. Bannon, J. K. and Steele, L. P. (1960) Average water-vapour content of the air. Geophysical Memoirs 102, Meteorological Office (38pp.). Borchert, J. R. (1971) The dust bowl in the 1970s. Ann. Assn Amer. Geogr. 61, 1–22. Browning, K. (1993) The global energy and water cycle. NERC News July, 21–3. Bryson, R. A. (1973) Drought in the Sahel: who or what is to blame? The Ecologist 3(10), 366–71. Chacon, R. E. and Fernandez, W. (1985) Temporal and spatial rainfall variability in the mountainous region of the Reventazon River Basin, Costa Rica. J. Climatology 5, 175–88. Chagnon, S. A. (2002) Frequency of heavy rainstorms on areas from 10 to 10,000km 2 , defined using dense rain gauge networks. J. Hydromet . 3(2), 220–3. Choudhury, B. J. (1993). Desertification, in

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5 Atmospheric instability, cloud formation and precipitation processes

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Books

Byers, H. R. and Braham, R. R. (1949) The Thunderstorm , US Weather Bureau. [Classic study of thunderstorm processes]

Cotton, W. R. and Anthes, R. A. (1989) Storm and Cloud Dynamics, Academic Press, San Diego, CA, 883pp. [Discusses cloud types and physical and dynamical processes, mesoscale structures, and the effects of mountains on airflow and cloud formation]

Kessler, E. (ed.) (1986) Thunderstorm Morphology and Dynamics , University of Oklahoma Press, Norman, OK, 411pp. [Comprehensive accounts by leading experts on convection and its modeling, all aspects of thunderstorm processes and occurrence in different environments, hail, lightning and tornadoes]

Ludlam, F. H. (1980) Clouds and Storms. The Behavior and Effect of Water in the Atmosphere , Pennsylvania State University, University Park and London, 405pp. [A monumental work by a renowned specialist]

Mason, B. J. (1975) Clouds, Rain and Rainmaking (2nd edn), Cambridge University Press, Cambridge and New York,189pp. [Valuable overview by a leading cloud physicist]

Petterssen, S. (1969) Introduction to Meteorology, 3rd edn, McGraw-Hill, New York, 416pp. Strahler, A. N. (1965) Introduction to Physical Geography , John Wiley & Sons, New York, 455pp. World Meteorological Organization (1956) International Cloud Atlas , Geneva. [Cloud classification and photographs of all sky types] Articles Andersson, T. (1980) Bergeron and the oreigenic (orographic) maxima of precipitation. Pure Appl. Geophys . 119, 558–76. Bennetts, D. A., McCallum, E. and Grant, J. R. (1986) Cumulonimbus clouds: an introductory review. Met. Mag. 115, 242–56. Bergeron, T. (1960) Problems and methods of rainfall investigation, in The Physics of Precipitation, Geophysical Monograph 5, Amer. Geophys. Union, Washington, DC, 5–30. Bering, E. A. III., Few, A. A. and Benbrook, J. R. (1998 The global electric circuit. Physics Today 51(9), 24–30. Braham, R. R. (1959) How does a raindrop grow? Science 129, 123–9. Browning, K. A. (1980) Local weather forecasting. Proc. Roy. Soc. Lond. Sect. A 371, 179–211.

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• Account for the differences between the environmental, dry adiabatic and saturated adiabatic lapse rates.

• What processes determine the presence of stability and instability in the troposphere?

• What factors cause air to ascend/descend on small and large scales and what are the associated weather outcomes?

• Maintain a record of cloud type and amount over several days and compare what you observe with the cloud cover shown for your location on satellite imagery from an appropriate website (see Appendix 4D).

• Make a cross-section of terrain height and precipitation amounts at stations along a height transect in your own region/country. Use daily, monthly or annual data as available. In addition, note the prevailing wind direction with respect to the mountains/hills.

• From national records/websites, examine the occurrence of convective systems (thunderstorms, tornadoes, lightning) in your country and determine whether they are air mass storms connected with frontal lows, or mesoscale convective systems. Chacón, R. E. and Fernandez, W. (1985) Temporal and spatial rainfall variability in the mountainous region of the Reventazón river basin,Costa Rica. Int. J. Climatol. 5, 176–88. Chinn, T. J. (1979) How wet is the wettest of the wet West Coast? New Zealand Alpine Journal 32, 85–7. Dudhia, J. (1996) Back to basics: thunderstorms. Part 1. Weather 51(11), 371–6. Dudhia, J. (1997) Back to basics: thunderstorms. Part 2, Storm types and associated weather. Weather 52(1), 2–7. Durbin, W. G. (1961) An introduction to cloud physics. Weather 16, 71–82 and 113–25. East, T. W. R. and Marshall, J. S. (1954) Turbulence in clouds as a factor in precipitation. Quart. J. Roy. Met. Soc. 80, 26–47. Eyre, L. A. (1992) How severe can a ‘severe thunderstorm’ be? Weather 47, 374–83. Griffiths, D. J., Colquhoun, J. R., Batt, K. L. and Casinader, T. R. (1993) Severe thunderstorms in New South Wales: climatology and means of assessing the impact of climate change. Climatic Change 25, 369–88. Henderson, R. (1993) Extreme storm rainfalls in the Southern Alps, New Zealand, in Extreme Hydrological Events: Precipitation,

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6 Atmospheric motion: principles

READING

Books

Barry, R. G. (2008) Mountain Weather and Climate, Cambridge University Press, 506 pp.) [Chapter on circulation systems related to orographic effects]

Oke, T. R. (1978) Boundary Layer Climates, Methuen, London, 372pp. [Prime text on surface climate processes in natural and humanmodified environments]

Scorer, R. S. (1958) Natural Aerodynamics, Pergamon Press, Oxford 312pp.

Simpson, J. E. (1994) Sea Breeze and Local Wind, Cambridge University Press, Cambridge, 234pp. [A well-illustrated descriptive account of the sea breeze and its effects; on e chapter on local orographic winds] Troen, I. and Petersen, E. L. (1989) European Wind Atlas ,Commissionof the Economic Community, Risø National Laboratory, Roskilde, Denmark, 656pp. Wells, N. (1986) The Atmosphere and Ocean. A Physical Introduction , Taylor & Francis, London 345pp. [Good account of both systems and their interactions] Articles Banta, R.M. (1995) Sea breezes: shallow and deep on the California coast. Mon. Wea. Rev. 123(12). 3614-22.

Air motion is described by its horizontal and vertical components; the latter are much smaller than

the horizontal velocities. Horizontal motions compensate for vertical imbalances between

gravitational acceleration and the vertical pressure gradient. The horizontal pressure gradient, the earth’s rotational effect (Coriolis force), and the curvature

of the isobars (centripetal acceleration) determine horizontal wind velocity. All three factors are

accounted for in the gradient wind equation, but this can be approximated in large-scale flow by

the geostrophic wind relationship. Below 1500m, the wind speed and direction are affected

by surface friction. Air ascends (descends) in association with surface convergence (divergence) of air. Air motion

is also subject to relative vertical vorticity as a result of curvature of the streamlines and/or lateral

shear; this, together with the earth's rotational effect, makes up the absolute vertical vorticity. Local winds occur as a result of diurnally varying thermal differences setting up local pressure

gradients (mountain–valley winds and land–sea breezes) or due to the effect of a topographic barrier

on airflow crossing it (examples are the lee-side föhn and bora winds).

• Compare the wind direction and speed reported at a station near you with the geostrophic wind velocity determined from the MSL pressure map for the same time (data sources are listed in Appendix 4).

• Why would there be no 'weather' if the winds were strictly geostrophic?

• What are the causes of mass divergence (convergence) and what roles do they play in weather processes?

• In what situations do local wind conditions differ markedly from those expected for a given largescale pressure gradient? Beran, W. D. (1967) Large amplitude lee waves and chinook winds. J. Appl. Met. 6, 865–77. Brinkmann, W. A. R. (1971) What is a foehn? Weather 26, 230–9. Brinkmann, W. A. R. (1974) Strong downslope winds at Boulder, Colorado. Monthly Weather Review 102, 592–602. Buettner, K. J. and Thyer, N. (1965) Valley winds in the Mount Rainer area. Archiv. Met. Geophys. Biokl. B 14, 125–47. Eddy, A. (1966) The Texas coast sea-breeze: a pilot study. Weather 21, 162–70. Ernst, J. A. (1976) SMS-1 night-time infrared imagery of low-level mountain waves. Monthly Weather Review 104, 207–9. Flohn, H. (1969) Local wind systems, in Flohn, H. (ed.) General Climatology, World Survey of Climatology 2, Elsevier, Amsterdam, 139–71. Galvin., J.F.P. (2007)The weather and climate of the tropics. Part 2 – The subtropical jet streams. Weather 62, 295–99. Geiger, R. (1969) Topoclimates, in Flohn, H. (ed.) General Climatology, World Survey of Climatology 2, Elsevier, Amsterdam, 105–38. Glenn, C. L. (1961) The chinook. Weatherwise 14, 175–82. Johnson, A. and O'Brien, J. J. (1973) A study of an Oregon sea breeze event. J. Appl. Met. 12, 1,267–83. Lockwood, J. G. (1962) Occurrence of föhn winds in the British Isles. Met. Mag. 91, 57–65.

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7 Planetary-scale motions in the atmosphere and ocean

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Bottomley, M., Folland, C. K., Hsiung, J., Newell, R. E. and Parker, D.E. (1990) Global Ocean Surface Temperature Atlas , Meteorological Office, London, 20pp + 313 plates.

Corby, G. A. (ed.) (1970) The Global Circulation of the Atmosphere , Roy. Met. Soc., London, 257pp.

Flohn, H. and Fantechi, R. (eds) (1984) The Climate of Europe: Past, Present and Future , D. Reidel, Dordrecht, 356pp.

Halitner, G. J. and Martin, F. L. (1957) Dynamical and Physical Meteorology , McGraw-Hill, New York, 470pp. [Classic text]

Houghton, J. (ed.) (1984) The Global Climate, Cambridge University Press, Cambridge, 233pp.

Indian Meteorological Department (1960) Monsoons of the World, Delhi, 270pp.

Kuenen, Ph. H. (1955) Realms of Water, CleaverHulme Press, London, 327pp.

Levitus, S. (1982) Climatological Atlas of the World Ocean , NOAA Professional Paper No. 13, Rockville, MD, 173pp.

Lorenz, E. N. (1967) The Nature and Theory of the General Circulation of the Atmosphere , World Meteorological Organization, Geneva, 161pp. [Classic account of the mechanisms and maintenance of the global circulation, transports of momentum, energy and moisture]

NASA (n.d.) From Pattern to Process: The Strategy of the Earth Observing System (Vol. II), EOS Science Steering Committee Report, NASA, Houston. Riley, D. and Spalton, L. (1981) World Weather and Climate (2nd edn), Cambridge University Press, London, 128pp. [Elementary text] Stowe, K. (1983) Ocean Science (2nd edn), John Wiley & Sons, New York, 673pp. Strahler, A. N. and Strahler, A. H. (1992)

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• What features of the global wind belts at the surface and in the upper troposphere are in accord with (differ from) those implied by the three-cell model of meridional circulation?

• What are the consequences of the westerly jet streams for transoceanic air travel?

• Examine the variation of the vertical structure of the zonal wind by creating height cross-sections for different longitudes and months using the CDC website (http://www.cdc.noaa.gov).

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8 Numerical models of the general circulation, climate and weather prediction

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9 Mid-latitude synoptic and mesoscale systems

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Petterssen, S. and Smeybe, S.J. (1971) On the development of extratropical cyclones. Quart. J. Roy. Met. Soc. 97, 457–82. scale features. The mid-latitude depression is usually associated with the convergence of contrasting air masses. According to the ‘Norwegian cyclone model’ (see Figure 9.7), the interface between these air masses develops into a wave form with its apex located at the center of the low pressure area. The wave encloses a mass of warm air between modified cold air in front and fresh, cold air in the rear. The formation of the wave also creates a distinction between the two sections of the original air-mass discontinuity since, although each section still marks the boundary between cold and warm air, the weather characteristics found within each section are very different. The two sections of the frontal surface are distinguished by the names warm front for the leading edge of the wave and the warm air and cold front for that of the cold air to the rear (see Figure 9.7B). The boundary between two adjacent air masses is marked by a strongly baroclinic zone of large temperature gradient, 100–200km wide (see C, this chapter, and Figure 9.1). Sharp discontinuities W W W W W W

C

C C C C

C C o o l

(A)

Early stage (B) Open stage

(C)

Occluded stage (D) Dissolving stage

Figure 9.7 Four stages in the typical develop

ment of a mid-latitude depression. Satellite views

of the cloud systems corresponding to these

stages are shown in Figure 9.8.

Source: Mostly after Strahler (1965), modified after

Beckinsale.

Notes: C = cold air; W = warm air.

(A) (B)

(C) (D) (E) Cloud covered Mostly open L L L L L

Figure 9.8 Schematic patterns of cloud cover (white) observed from satellites, in relation to surface

fronts and generalized isobars. A, B, C and D correspond to the four stages in Figure 9.7.

Source: After Boucher and Newcomb (1962). Courtesy of the American Meteorological Society. of temperature, moisture and wind properties at fronts, especially the warm front, are rather uncommon. Such discontinuities are usually the result of a pronounced surge of fresh, cold air in the rear sector of a depression, but in the middle and upper troposphere they are often caused by subsidence and may not coincide with the location of the baroclinic zone. In meteorological analysis centers, numerous criteria are used to locate frontal boundaries: 1000–500mb thickness gradients, 850mb wet-bulb potential temperature, cloud and precipitation bands, and wind shifts. However, a forecaster may have to use judgement when some of these criteria disagree. On satellite imagery, active cold fronts in a strong baroclinic zone commonly show marked spiral cloud bands, formed as a result of the thermal advection (Figure 9.8B, C). A cirrus shield, however, typically covers warm fronts. As Figure 9.7 shows, an upper tropospheric jet stream is closely associated with the baroclinic zone, blowing roughly parallel to the line of the upper front. This relationship is examined below. Air behind the cold front, away from the low center, commonly has an anticyclonic trajectory and hence moves at a greater than geostrophic speed (see Chapter 5A.4), impelling the cold front to acquire a supergeostrophic speed also. The L I II III IV I II III IV L L L (A) Pressure, fronts and clouds (B) Temperature and air currents WARM Figure 9.9 Stages in the life-cycle of a marine extratropical depression showing I: incipient frontal depression, II: frontal fracture, III: bent-back warm front and frontal T-bone, IV: warm-core seclusion. A: Schematic isobars of sea-level pressure, fronts and cloud cover (stippled). B: Isotherms

and flows of cold air (solid arrows) and warm air (dashed arrows). Source: After Shapiro and Keyser (1990). By permission of the American Meteorological Society.

wedge of warm air is pinched out at the surface and lifted bodily off the ground. This stage of occlusion eliminates the wave form at the surface (see Figure 9.7). The depression usually achieves its maximum intensity 12–24 hours after the beginning of occlusion. The occlusion gradually works outwards from the center of the depression along the warm front. Sometimes, the cold air wedge advances so rapidly that, in the friction layer close to the surface, cold air overruns the warm air and generates a squall line (see Chapter 4G).

By no means all frontal lows follow the idealized life-cycle discussed above. It is generally characteristic of oceanic cyclogenesis, although the evolution of those systems has been re examined using aircraft observations collected during North Atlantic meteorological field programs in the 1980s. These suggest a different evolution of maritime frontal cyclones (Figure 9.9). Four stages are identified: (1) cyclone inception features a broad (400km) continuous frontal zone; (2) frontal fracture occurs near

the center of the low with tighter frontal gradients; (3) a T-bone structure and bent-back warm front develop; and (4) the mature cyclone shows seclusion of the warm core within the polar airstream behind the cold front.

Over central North America, cyclones forming in winter and spring depart considerably from the Norwegian model. They often feature an outflow of cold arctic air east of the Rocky Mountains forming an arctic front, a lee trough with dry air descending from the mountains, and warm, moist southerly flow from the Gulf of Mexico (Figure 9.10). The trough superposes dry air over warm, moist air, generating instability and a rain band analogous to a warm front. The arctic air moves southward west of the low center, causing lifting of warmer, dry air but giving little precipitation. There may also be an upper cold front advancing over the trough that forms a rain band along its leading edge. Such a system is thought to have caused a record rainstorm at Holt, Missouri, on 22 June 1947, when 305mm fell in just 42 minutes! Gulf of Mexico (A) Gulf of Mexico (B) (C) CFA CFA CFA Potentially unstable region Dry trough Dry trough Dry trough CFA rain band at surface Upward motion Upward motion Low-θ e Air High-θ e Air Figure 9.10 Schematic model of a dry trough and frontogenesis east of the Rocky Mountains. A: Warm, dry air with low equivalent potential temperature (u e) from the Rockies overrides warm, moist, high u e air from the

Gulf of Mexico, forming a potentially unstable zone east of the dry trough. B: Upward motion associated with the cold front aloft (CFA). C: Location of the CFA rain band at the surface. [Equivalent potential temperature is the potential temperature of an air parcel that is expanded adiabatically until all water vapor is condensed and the latent heat released then compressed adiabatically to 1000mb pressure.] Source: After Locatelli et al. (1995). By permission of the American Meteorological Society. E FRONTAL CHARACTERISTICS The character of frontal weather depends upon the vertical motion in the air masses. If the air in the warm sector is rising relative to the frontal zone the fronts are usually very active and are termed ana-fronts, whereas sinking of the warm air relative to the cold air masses gives rise to less intense kata-fronts (Figure 9.11). 1 The warm front The warm front represents the leading edge of the warm sector in the wave. The frontal boundary has a very gentle slope, of the order of 0.5–1°, so the cloud systems associated with the upper portion of the front herald its approach some 12 hours or more before the arrival of the surface front. The ana-warm front, with rising warm air, has multi-layered cloud that steadily thickens and lowers towards the surface position of the front. The first clouds are thin, wispy cirrus, followed by sheets of cirrus and cirrostratus, and altostratus (Figure 9.11A). The sun is obscured as the altostratus layer thickens and drizzle or rain begins to fall. The cloud often extends through most of the troposphere and, with continuous precipita0500 Rain belt Dry descent Rain belt 1000 Movement of system (A) (B) COLD AIR COLD AIR COLD AIR COLD AIR Cu Ci Ci Cs Cs convective layer top F r o n t a l z o n e F r o n t a l F r o n t a l z o n e F r o n t a l z o n e z o n e As Cu St Ns Sc Sc and Cu Sc Sc Cu Sc St Thick Thick J L J U WARM AIR WARM AIR Base of subsidence inversion Movement of system 15002000 0 5000 10000 Height (m) kilometers 050010001500 Rain belt Belt of light rain or drizzle 2000 0 5000 10000 Height (m) kilometers T R O P O P A U S E T R O P O P A U S E T R O P O P A U S E T R O P O P A U S E Figure 9.11 A: Cross-sectional model of a depression with ana-fronts, where the air is rising relative to each frontal surface. Note that an ana-warm front may occur with a kata-cold front and vice versa. JU and JL show the locations of the upper and lower jet streams, respectively. B: Model of a depression with katafronts, where the air is sinking relative to each frontal surface. Sources: After Pedgley, A Course in Elementary Meteorology, and Bennetts et al. (1988). (Crown copyright ©), reproduced by permission of the Controller of Her Majesty’s Stationery Office.

tion occurring, is generally designated as nimbostratus. Patches of fractostratus may also form in the cold air as rain falling through this air undergoes evaporation and quickly saturates it.

The descending warm air of the kata-warm front greatly restricts the development of medium- and high-level clouds. The frontal cloud is mainly stratocumulus, with a limited depth as a result of the subsidence inversions in both air masses (see Figure 9.11B). Precipitation is usually light rain or drizzle formed by coalescence.

At the passage of the warm front the wind veers, the temperature rises and the fall of pressure is checked. The rain becomes intermittent or ceases in the warm air and the thin stratocumulus cloud sheet may break up.

Forecasting the extent of rain belts associated with the warm front is complicated by the fact that most fronts are not ana- or kata-fronts throughout their length or even at all levels in the troposphere. For this reason, radar is increasingly being used to map the precise extent of rain belts and to detect differences in rainfall intensity. Such studies show that most of the production and distribution of precipitation is controlled by a broad airflow a few hundred kilometers across and several kilometers deep, which flows parallel to and ahead of the surface cold front (see Figure 9.12). Just ahead of the cold front, the flow occurs as a low-level jet with winds of up to 25–30m s −1 at about 1km above the surface. The air, which is warm and moist, rises over the warm front and turns southeastward ahead of the front, merging with the mid-tropospheric flow (B in Figure 9.13). This flow is termed a ‘conveyor belt’ (for large-scale C C C

B B B A A A 9 0 0 m b

4 0 0 m b 5 0 0 m b 5 0 0 m b 7 0 0 m b

6 0 0 m b 9 0 0 m b 0 200 km Conveyor belt of ascending air Mid-troposphere flow Subsiding flow ahead of occluded front Precipitation (including bels of higher intensity)

Figure 9.12 Model of the large-scale flow and

mesoscale precipitation structure of a partially

occluded depression typical of those affecting the

British Isles. It shows the ‘conveyor belt’ (A) rising

from 900mb ahead of the cold front over the warm

front. This is overlaid by a mid-tropospheric flow (B)

of potentially colder air from behind the cold front.

Most of the precipitation occurs in the well-defined

region shown, within which it exhibits a cellular and

banded structure.

Source: After Harrold (1973). By permission of the Royal

Meteorological Society. Cirrus cloud boundary Low 3b 3b 2 1b SYNOPTIC FEATURES TYPES OF MESOSCALE RAINBANDS 1 2 3 4 5 Warm frontal Warm sector Cold frontal Prefrontal cold surge Postfrontal Surface Cold front Cold front Prefrontal cold surge Warm front Warm occluded front Aloft 1a N Y X 3a 0 300 km 3a 5 5 4b 4a Figure 9.13 Fronts and associated rain bands typical of a mature depression. The broken line X–Y shows the location of the cross-section given in Figure 9.14. Source: After Hobbs; from Houze and Hobbs (1982). By permission of Academic Press. heat and momentum transfer in mid-latitudes). Broad-scale convective (potential) instability is generated by the overrunning of this low-level flow by potentially colder, drier air in the middle troposphere. Instability is released mainly in small-scale convection cells that are organized into clusters, known as mesoscale precipitation areas (MPAs). These MPAs are further arranged in bands, 50–100km wide (Figure 9.13). Ahead of the warm front, the bands are broadly parallel to the airflow in the rising section of

the conveyor belt, whereas in the warm sector they parallel the cold front and the low-level jet. In some cases, cells and clusters are further arranged in bands within the warm sector and ahead of the warm front (see Figures 9.13 and 9.14). Precipitation from warm front rain bands often involves ‘seeding’ by ice particles falling from the upper cloud layers. It has been estimated that 20–35 percent of the precipitation originates in the ‘seeder’ zone and the remainder in the lower clouds (see also Figure 5.14). Orographic effects set up some of the cells and clusters, and these may travel downwind when the atmosphere is unstable. 2 The cold front The weather conditions observed at cold fronts are equally variable, depending upon the stability of the warm-sector air and the vertical motion relative to the frontal zone. The classical coldfront model is of the ana-type, and the cloud is usually cumulonimbus. Figure 9.15 illustrates the warm conveyor belt associated with such a frontal zone and the line convection. Over the British Isles, air in the warm sector is rarely unstable, so that nimbostratus occurs more frequently at the cold front (see Figure 9.11A). With the kata-cold front the cloud is generally stratocumulus (see Figure 9.11B) and precipitation is light. With ana-cold fronts there are usually brief, heavy downpours, sometimes accompanied by thunder. The steep slope of the cold front, roughly 2°, means that the bad weather is of shorter duration than at the warm front. With the passage of the cold front the wind veers sharply, pressure begins to rise and temperature falls. The sky may clear abruptly, even before the passage of the surface cold front in some cases, although with kata-cold fronts the changes are more gradual. Forward-tilting cold fronts are sometimes observed, either due to surface friction – especially an orographic barrier – slowing the low-level motion of the front, or as a result of a cold front aloft (see Figure 9.10). 3 The occlusion The cold front moves faster relative to the warm front, eventually catching up with it, leading to

X Y REAR OF CLOUD SHIELD EVAPORATION CUMULONIMBUS TOWERS DRY [3-50] [ ] [2-16] [0.5-4] [30140] (0.1) (0.4) (0.3) (0.2) (0.05) [4-16] STABLE STRATAFORM CLOUDS GROUND 0°C (0.4 1.4)

Water particles ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ ∗ Ice particles Ice particle concentration (per litre) Embedded convection cells Cloud liquid water content (g m -3) ( )

15

10

5 5 3a 3a 2 1b 1a km 0 100 3b

0

km

Figure 9.14 Cross-section along the line X–Y in Figure 9.13 showing cloud structures and rain bands. The vertical

hatching represents rainfall location and intensity. Raindrop and ice particle regions are shown, as are ice particle

concentrations and cloud liquid water content. Numbered belts refer to those shown in Figure 9.13. Scales are

approximate.

Source: After Hobbs and Matejka et al.; from Houze and Hobbs (1982). By permission of Academic Press.

occlusion, where the warm sector air is lifted off the

ground. Occlusions are classified as either cold or

warm, depending on the relative states of the air

masses lying in front and to the rear of the warm

sector (Figure 9.16). If air mass 2 is colder than air

mass 1, then the occlusion is warm, but if the

reverse is so it is termed a cold occlusion. The air

in advance of the depression is likely to be coldest

when depressions occlude over Europe in winter

and very cold cP air is affecting the continent.

Recent work suggests that most occlusions are

warm and that the thermal definition is often

misleading. A new definition is proposed: a cold

(warm) occlusion forms when the air that is more

statically stable lies behind the cold front (ahead

of the warm front) (Figure 9.16).

The line of the warm air wedge aloft is

associated with a zone of layered cloud (similar

to that found with a warm front) and often of

precipitation. Hence its position is indicated

separately on some weather maps and it is referred

to by Canadian meteorologists as a trowal (trough

of warm air aloft). The passage of an occluded front and trowal brings a change back to polar airmass weather. A different process occurs when there is interaction between the cloud bands within a polar trough and the main polar front, giving rise to an instant occlusion. A warm conveyor belt on the polar front ascends as an upper tropospheric jet, forming a stratiform cloud band (Figure 9.17), while a low-level polar trough conveyor belt at right angles to it produces a convective cloud band and precipitation area poleward of the main polar front on the leading edge of the cold pool. Frontolysis (frontal decay) represents the final phase of a front's existence although it is not necessarily linked with occlusion. Decay occurs when differences no longer exist between adjacent air masses. This may arise in four ways: (1) through their mutual stagnation over a similar surface; (2) as a result of both air masses moving on parallel tracks at the same speed; (3) through their movement in succession along the same track at the same speed; or (4) by the system entraining air of the same temperature.

Descending dry air with low θ w Warm conveyor belt 6km 3km Cold-front zone 40cm/s up 5m/s up Lowlevel jet Surface cold front XY 20cm/s down Line convection Y X

(A) (B)

Figure 9.15 Schematic diagrams showing airflows, relative to the moving frontal system, at an ana-cold front. A

warm conveyor belt (stippled) ascends above the front with cold air (dashed arrows) descending beneath it. A: Plan

view. B: Vertical section along the line X–Y, showing rates

of vertical motion.

Source: Browning (1990). By permission of the American Meteorological Society. B A D C Air mass 2 Air mass 2: Colder Air mass 2: Cold Air mass 2 Air mass 1: Colder Air mass 1: Cold A B C D A B C D Air mass 1 Air mass 1: More statically stable Air mass 2: More statically stable Warm sector air Warm sector air Warm sector air Warm sector air Air mass 2 Air mass 1 Air mass 1 Warm occlusion Cold occlusion L L Figure 9.16 Schematic illustration of a cold and a warm occlusion in the classical model. Plan view of synoptic pattern (above) and cross-sections along line A–B (center). Colder air is shaded darker. The bottom panel illustrates proposed criteria for identifying warm and cold occlusions based on static stability. Source: Above and center from Stoelinga et al., (2002, p. 710, Fig. 1). Courtesy of the American Meteorological Society. The bottom panel is based on their new definition.

4 Frontal-wave families

Observations show that frontal waves over the

oceans, at least, do not generally occur as separate

units but in families of three or four (see Figure 9.9). The depressions that succeed the original one form as secondary lows along the trailing edge of an extended cold front. Each new member follows a course that is south of its progenitor as the polar air pushes further south to the rear of COMMA DEVELOPMENT CCB CCB L A T E R T I M E E A R L I E R CCB CCB D CCB W C B 2 W C B 2 W C B W C B 1 W C B 1 W C B W C B D R Y A I R WCB1 WCB1 DRY AIR D R Y A I R INSTANT OCCLUSION FRONTAL-WAVE SEQUENCE

Figure 9.17 Schematic illustrations of vortex developments in satellite imagery. The sequences run from

bottom to top. Left: comma cloud (C) developing in a polar airstream. Center: instant occlusion from the

interaction of a polar trough with a wave on the polar front. Right: the classical frontal wave with cold and

warm conveyor belts (CCB, WCB). C = enhanced convection; D = decaying cloud band; cloud cover

stippled.

Source: After Browning (1990). By permission of the

American Meteorological Society. each depression in the series. Eventually, the front trails far to the south and the cold polar air forms an extensive meridional wedge of high pressure, terminating the sequence. Another pattern of development may take place on the warm front, particularly at the point of occlusion, as a separate wave forms and runs ahead of the parent depression. This type of secondary is more likely with very cold (cA, mA or cP) air ahead of the warm front, and it tends to form when mountains bar the eastward movement of the occlusion. This situation often occurs when a primary depression is situated in the Davis Strait and a breakaway wave forms south of Cape Farewell (the southern tip of Greenland), moving away eastward. Analogous developments take place in the Skagerrak–Kattegat area when the occlusion is held up by the Scandinavian mountains. F ZONES OF WAVE DEVELOPMENT AND FRONTOGENESIS Fronts and associated depressions tend to develop in well-defined areas. The major zones of frontal wave development are areas that are most frequently baroclinic as a result of airstream confluence (Figure 9.18). This is the case off East Asia and eastern North America, especially in winter, when there is a sharp temperature gradient between the snow-covered land and warm offshore currents. These zones are referred to as the Pacific Polar and Atlantic Polar Fronts, respectively (Figure 9.19). Their position is quite variable, but they are displaced equatorward in winter, when the Atlantic Frontal Zone may extend into the Gulf of Mexico. Here there is convergence of air masses of different stability between adjacent subtropical high pressure cells. Depressions developing here commonly move northeastward, sometimes following or amalgamating with others of the northern part of the Polar Front proper or of the Canadian Arctic Front. Frontal frequency remains high across the North Atlantic, but it decreases eastward in the North Pacific, perhaps owing to a less pronounced gradient of sea surface temperature. Frontal activity is most common in the central North Pacific when the subtropical high is split into two cells with converging airflows between them. Another section of the Polar Front, often referred to as the Mediterranean Front, is located over the Mediterranean–Caspian Sea areas in winter. At intervals, fresh Atlantic mP air, or cool cP air from southeast Europe, converges with warmer air masses of North African origin over the Mediterranean Basin and initiates frontogenesis. In summer, the Azores subtropical anticyclone influences the area, and the frontal zone is absent. The summer locations of the Polar Front over the western Atlantic and Pacific are some 10° further north than in winter (see Figure 9.19), although the summer

frontal zone is rather weak. There is a frontal zone over Eurasia and a corresponding one over middle North America. These reflect the general meridional temperature gradient and also the large-scale influence of orography on the general circulation (see G, this chapter). In the Southern Hemisphere, the Polar Front is on average about 45°S in January (summer), with branches spiralling poleward towards it from about 32°S off eastern South America and from 30°S, 150°W in the South Pacific (Figure 9.20). In July (winter), there are two Polar Frontal Zones spiralling towards Antarctica from about 20°S; one starts over South America and the other at 170°W. They terminate some 4–5° latitude further poleward than in summer. It is noteworthy that the Southern Hemisphere has more cyclonic activity in summer than does the Northern Hemisphere in its summer. This appears to be related to the seasonal strengthening of the meridional temperature gradient (see p. 180). The second major frontal zone is the Arctic Front, associated with the snow and ice margins of high latitudes (see Figure 9.19). In summer, this zone is developed at the land–sea boundary in 180° January L L L L L L L L L L L L L H H H H H L L H H H H H L H H H L L H H H H H P F A F A F A F A F I T C Z I T C Z I T C Z I T C Z S P C Z

S P C Z S P C Z P F P F P F A F M T M T M T Z A B Z A B P F A A F A A F A A F P F P F P F P F P F P F P F P F A A F I T C Z M F P F M T P F I T C Z S P C Z A F 180°120° 120°60°W 60°E0°

0° 0°

20°N

20°S

40°

40°

60° 1 0 2 0 1 0 2 0 1 0 1 5 1 0 2 0 1 0 2 0 1 0 1 5 1 0 1 5 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 5 1 0 0 0 9 9 5 9 9 0 9 9 0 9 8 5 9 8 5 1 0 1 5 1 0 1 5 1 0 1 5 1 0 1 5 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 7 . 5 1 0 1 5 1 0 0 5 1 0 0 0 9 9 5 1 0 3 0 1 0 3 5 1 0 1 0 1 0 0 0 1 0 2 0 1 0 2 0 1 0 2 0 1 0 2 0 1 0 2 0 1 0 2 5 1 0 2 0 1 0 1 5 1 0 1 0 1 0 0 5 9 9 5 9 9 0 9 8 5 9 9 0 9 9 5 9 8 5 1 0 0 0 1 0 2 0 1 0 2 0 1 0 1 5 1 0 1 5 1 0 1 0 1 0 0 5 1 0 0 0 9 9 5 9 9 0 9 8 5 1 0 2 5 1 0 2 5 9 9 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 5 1 0 1 5 1 0 2 0 1 0 0 5 1 0 0 0 1 0 1 5

1 0 1 5 1 0 1 0 9 9 5 1 0 0 5

60° 20°N 20°S 40° 40° 60° 60° 180° 180°120° 120°60°W 60°E0° 180° July 180°120° 120°60°W 60°E0°

0° 0°

20°N

20°S

40°

40°

60°

60° 20°N 20°S 40° 40° 60° 60° 180° 180°120° 120°60°W 60°E0°

Figure 9.18 Mean pressure (mb) and surface winds for the world in January and July. The major frontal

and convergence zones are shown as follows: Intertropical Convergence Zone (ITCZ), South Pacific

Convergence Zone (SPCZ), Monsoon Trough (MT), Zaire Air Boundary (ZAB), Mediterranean Front (MF),

Northern and Southern Hemisphere Polar Fronts (PF), Arctic Fronts (AF) and Antarctic Fronts (AAF).

Source: Partly from Liljequist (1970). Siberia and North America where there is a strong temperature gradient between the heated snowfree land and the largely sea ice-covered, cold Arctic Ocean. In winter over North America, it is formed between cA (or cP) air and Pacific maritime air modified by crossing the Coast Ranges and the Rocky Mountains). There is also a less pronounced Arctic Frontal Zone in the North Atlantic–Norwegian Sea area, extending along the Siberian coast. A similar weak frontal zone is found in winter in the Southern Hemisphere. It is located at 65–70°S near the edge of the Antarctic pack ice in the Pacific sector (see Figure 9.20), although few cyclones form there. Zones of airstream confluence in the Southern Hemisphere (cf. Figures 9.2B and 9.4B) are fewer and more persistent, particularly in coastal regions, than in the Northern Hemisphere. The principal tracks of depressions in the Northern Hemisphere in January are shown in Figure 9.21. The major tracks reflect the primary frontal zones discussed above. In summer, the Mediterranean

route is absent and lows move across Siberia; the other tracks are similar, although more zonal and located in higher latitudes (around 60°N). Between the two hemispherical belts of subtropical high pressure there is a further major convergence zone, the Intertropical Convergence Zone (ITCZ). Formerly this was designated as the Intertropical Front (ITF), but air-mass contrasts are not typical. The ITCZ moves seasonally away from the equator, as the subtropical high pressure cell activity alternates in opposite hemispheres.

90°W 90°E 0° 180° E q u a t o r i a l Z o n e 3 0 ° S 6 0 ° S P O L A R F R O N T (W i) P O L A R F R O N T (S u) A N T A R C T I C F R O N T (W i)

Figure 9.20 The major Southern Hemisphere frontal zones

in winter (Wi) and summer (Su). Winter 3 0 ° N 6 0 ° N Summer 3 0 ° N 6 0 ° N P A C I F I C P O L A R F R O N T P A C I F I C P O L A R F R O N T A T L A N T I C P O L A R F R O N T A T L A N T I C P O L A R F R O N T S I B E R I A N C A N A D I A N A R C T I C F R O N T E U R A S I A N P O L A R F R O N T M E D I T E R R A N E A N F R O N T A T L A N T I C A S I A T I C A R C T I C F R O N T C A N A D I A N A R C T I C F R O N T Figure 9.19 The major Northern Hemisphere frontal zones in winter and summer.

The contrast between the converging air masses

obviously increases with the distance of the ITCZ

from the equator, and the degree of difference in

their characteristics is associated with considerable

variation in weather activity along the conver

gence zone. Activity is most intense in June to

July over South Asia and West Africa, when the

contrast between the humid maritime and dry

continental air masses is at a maximum. In these

sectors, the term Intertropical Front is applicable,

although this does not imply that it behaves

like a mid-latitude frontal zone. The nature

and significance of the ITCZ are discussed in

Chapter 11.

G SURFACE–UPPER-AIR

RELATIONSHIPS AND THE

FORMATION OF FRONTAL

CYCLONES

We have noted that a wave depression is associ

ated with air-mass convergence, yet the baro

metric pressure at the center of the low may 90°W 90°E 0° 180° E q u a t o r i a l Z o n e 3 0 ° N 6 0 ° N Figure 9.21 The principal Northern Hemisphere depression tracks in January. The full lines show major tracks, the dashed lines secondary tracks that are less frequent and less well defined. The frequency of lows is a local maximum where arrowheads end. An area of frequent cyclogenesis is indicated where a secondary track changes to a primary track or where two secondary tracks merge to form a primary track. Source: After Klein (1957). Courtesy of the US Weather Bureau.

Plate 9.1 A composite image mostly from the Moderate resolution Imaging Spectrometer (MODIS) on NASA's

Terra satellite orbiting 700km above the earth.

The cloud cover is a composite of visible and thermal infrared imagery for 29 July and November 2001. City

lights are superimposed from Defense Meterological Satellite Program observations over a nine-month period.

Topographic shading is from the US Geological Survey GTOPO 30 dataset.

Source: Blue Marble Visible Earth, NASA ftp://gloria 2-f.gsfc.nasa.gov/pub/stockli. decrease by 10–20mb in 12–24 hours as the system intensifies. This is possible because upper-air divergence removes rising air more quickly than convergence at lower levels replaces it (see Figure 5.7). The superimposition of a region of upper divergence over a frontal zone is the prime motivating force of cyclogenesis

(i.e., depression formation). The long (or Rossby) waves in the middle and upper troposphere, discussed in Chapter 7A.2, are particularly important in this respect. The latitudinal circumference limits the circumpolar westerly flow to between three and six major Rossby waves, and these affect the formation and movement of surface depressions. Two primary stationary waves tend to be located about 70°W and 150°E in response to the influence on the atmospheric circulation of orographic barriers, such as the Rocky Mountains and the Tibetan plateau, and of heat sources. On the eastern limb of troughs in the upper westerlies of the Northern Hemisphere the flow is normally divergent, since the gradient wind is subgeostrophic in the trough but supergeostrophic in the ridge (see Chapter 6A.4). Thus, the sector ahead of an upper trough is a very favorable location for a surface depression to form or deepen (see Figure 9.22). It will be noted that the mean upper troughs are significantly positioned just west of the Atlantic and Pacific Polar Front Zones in winter. With these ideas in mind, we can examine the three-dimensional nature of depression development and the links existing between upper and lower tropospheric flow. The basic theory relates to the vorticity equation, which states that, for frictionless horizontal motion, the rate of change of the vertical component of absolute vorticity (dQ/dt or d(f + ζ)/dt) is proportional to air-mass convergence (−D, i.e., negative divergence): dQ 1 dQ --- = DQ or D = − -- --- dt Q dt The relationship implies that a converging (diverging) air column has increasing (decreasing) absolute vorticity. The conservation of vorticity equation, which we have already discussed, is in fact a special case of this relationship. In the sector ahead of an upper trough, the decreasing cyclonic vorticity causes divergence (i.e., D positive), since the change in ζ outweighs that in f, thereby favoring surface convergence and low-level cyclonic vorticity (see Figure 9.23). Once the surface cyclonic circulation has become established, vorticity production is increased through the effects of thermal advection. Poleward transport of warm air in the warm sector and 9km 9km 300mb 300mb R I D G E R I D G E T R O U G H D D D VN VN + VX VX V N + + C C C S S R R R R SURFACE Surface isobars Rossby waves at 300mb Vertical motions Rising Sinking Convergence Divergence Maximum vorticity Minimum vorticity Positive and negative vorticity SURFACE H L Figure 9.22 Schematic representation of the relationship between surface pressure (H and L), airflow and frontal systems, on the one hand, and the location of troughs and ridges in the Rossby waves at the 300mb level. The locations of maximum (cyclonic) and minimum (anticyclonic) relative vorticity are shown, as are those of negative (anticyclonic) and positive (cyclonic)

vorticity advection. Sources: Mostly after Musk (1988), with additions from Uccellini (1990). Courtesy of Cambridge University Press.

CON CON Surface front 300mb JET STREAM DIV DIV

S

I N

K I N G S I N K I N G v e l o c i t y v e l o c i t y m a x i m u m m a x i m u m R I S I N G

Figure 9.23 Model of the jet stream and surface fronts,

showing zones of upper tropospheric divergence and

convergence and the jet stream cores.

the eastward advance of the cold upper trough act

to sharpen the baroclinic zone, strengthening the

upper jet stream through the thermal wind

mechanism (see p. 170). The vertical relationship

between jet stream and front has already been

shown (see Figure 7.8); a model depression

sequence is demonstrated in Figure 9.23. The

actual relationship may depart from this idealized

case, although the jet is commonly located in the

cold air. Velocity maxima (core zones) occur

along the jet stream and the distribution of vertical

motion upstream and downstream of these cores

is known to be quite different. In the area of the

jet entrance (i.e., upstream of the core), divergence

causes lower-level air to rise on the equatorward

(i.e., right) side of the jet, whereas in the exit

zone (downstream of the core) ascent is on the

poleward side. Figure 9.24 shows how precipita

tion is more often related to the position of the jet

stream than to that of surface fronts; maximum

precipitation areas are in the right entrance sector

of the jet core. This vertical motion pattern is also of basic importance in the initial deepening stage of the depression. If the upper-air pattern is unfavorable (e.g., beneath left entrance and right exit zones, where there is convergence) the depression will fill. The development of a depression may also be considered in terms of energy transfers. A cyclone requires the conversion of potential into kinetic energy. The upward (and poleward) motion of warm air achieves this. The vertical wind shear and the superimposition of upper tropospheric divergence drive the rising warm air over a baroclinic zone. Intensification of this zone further strengthens the upper winds. The upper divergence allows surface convergence and pressure fall to occur simultaneously. Modern theory relegates the fronts to a subordinate role. They develop within depressions as narrow zones of intensified ascent, probably through the effects of cloud formation. Recent research has identified a category of mid-latitude cyclones that develop and intensify 100°

40° 40° 30° 20 Sept 1958 21 Sept 1958 > 25mm 1 0 1 2 1 0 1 2 1 0 1 2 1 0 1 2 1 0 1 2 1 0 1 2 1 0 2 0 1 0 2 0 1 0 0 4 1 0 0 4 1 0 0 4 1 0 0 4 9 9 6 H H H H L L L L > 45m s -1 ≤ 25mm

30°

110° 100° 90° 80° 100° 40° 40° 30° 30° 110° 100° 90° 80°

Figure 9.24 The relations between surface fronts and isobars, surface precipitation (≥25mm vertical hatching;

>25mm cross-hatching), and jet streams (wind speeds in excess of about 45m s −1 shown by stipple) over the United

States on 20 September 1958 and 21 September 1958. This illustrates how the surface precipitation area is related

more to the position of the jets than to that of the surface fronts. The air over the south-central United

States was

close to saturation, whereas that associated with the northern jet and the maritime front was much less moist.

Source: After Richter and Dahl (1958). Courtesy of the American Meteorological Society. rapidly, acquiring characteristics that resemble tropical hurricanes. These have been termed ‘bombs’ in view of their explosive rate of deepening; pressure falls of at least 24mb/24hr are observed. For example, the ‘QE II storm’, which battered the ocean liner Queen Elizabeth II off New York on 10 September 1978, developed a central pressure below 950mb with hurricaneforce winds and an eye-like storm center within 24 hours (see Chapter 11B.2). These systems are observed mainly during the cold season off the East Coast of the United States, off Japan, and over parts of the central and northeastern North Pacific, in association with major baroclinic zones and close to strong gradients of sea surface temperature. Explosive cyclogenesis is favored by an unstable lower troposphere and is often located downstream of a traveling 500mb-level trough. Bombs are characterized by strong vertical motion, associated with a sharply defined level of non-divergence near 500mb, and large-scale release of latent heat. Wind maxima in the upper troposphere, organized as jet streaks, serve to amplify the lower level instability and upward motion. Studies reveal that average cyclonic deepening rates over the North Atlantic and North Pacific are about 10mb/24hr, or three times greater than over the continental United States (3mb/24hr). Hence, it is suggested that explosive cyclogenesis represents a more intense version of typical maritime cyclone development. The movement of depressions is determined essentially by the upper westerlies and, as a rule of thumb, a depression center travels at about 70 percent of the surface geostrophic wind speed in the warm sector. Records for the United States indicate that the average speed of depressions is 32km hr −1 in summer and 48km hr −1 in winter. The higher speed in winter reflects the stronger westerly circulation. Shallow depressions are mainly steered by the direction of the thermal wind in the warm sector and hence their path closely follows that of the upper jet stream (see Chapter 6A.3). Deep depressions may greatly distort the thermal pattern, however, as a result of the northward transport of warm air and the southward transport of cold air. In such cases, the depression usually becomes slow moving. The movement of a depression may be additionally guided by energy sources such as a warm sea surface that generates cyclonic vorticity, or by mountain barriers. The depression may

cross obstacles, such as the Rocky Mountains or the Greenland Ice Sheet, as an upper low or trough, and subsequently redevelop, aided by the lee effects of the barrier or by fresh injections of contrasting air masses. Ocean surface temperatures can critically influence the location and intensity of storm tracks. Figure 9.25B indicates that an extensive, relatively warm surface in the north-central Pacific in the winter of 1971–1972 caused a northward displacement of the westerly jet stream together with a compensating southward displacement over the western United States, bringing in cold air there. This pattern contrasts with that observed during the 1960s (see Figure 9.25A), when a persistent cold anomaly in the central Pacific, with warmer water to the east, led to frequent storm development in the intervening zone of strong temperature gradient. The associated upper airflow produced a ridge over western North America with warm winters in California and Oregon. Models of the global atmospheric circulation support the view that persistent anomalies of sea surface temperature exert an important control on local and large-scale weather conditions. H NON-FRONTAL DEPRESSIONS Not all depressions originate as frontal waves. Tropical depressions are indeed mainly nonfrontal and these are considered in Chapter 11. In middle and high latitudes, four types that develop in distinctly different situations are of particular importance and interest: the lee cyclone, the thermal low, the polar low, and the cold low. 80° Cold pool Cold pool Cold Cold Warm pool Warm pool Warm Warm 70° 60° 50° 40° 30° 20°

140°

160° 180° 160° 140° 120° 100°

(A) Winter of 1960s 80° 70° 60° 50° 40° 30° 20°

140°

160° 180° 160° 140° 120° 100°

(B) Winter of 1971-72 J E T S T R E A M J E T S T R E A M S T O R M D E V E L O P M E N T Z O N E

Figure 9.25 Generalized relationships between ocean surface temperatures, jet stream tracks, storm

development zones and land temperatures over the North Pacific and North America during (A) average

winter conditions in the 1960s, and (B) the winter of

1971–1972, as determined by J. Namias.

Source: After Wick (1973). By permission of New Scientist. 1 The lee cyclone Westerly airflow that is forced over a north–south mountain barrier undergoes vertical contraction over the ridge and expansion on the lee side. This vertical movement creates compensating lateral expansion and contraction, respectively. Hence there is a tendency for divergence and anticyclonic curvature over the crest, and convergence and cyclonic curvature in the lee of the barrier. Wave troughs may be set up in this way on the lee side of low hills (see Figure 6.13) as well as major mountain chains like the Rocky Mountains. The airflow characteristics and the size of the barrier determine whether or not a closed low pressure system actually develops. Such depressions, which at least initially tend to remain ‘anchored’ by the barrier, are frequent in winter to the south of the Alps, when the mountains block the low-level flow of northwesterly airstreams. Fronts often develop in these depressions, but the low does not form as a wave along a frontal zone. Lee cyclogenesis is common in Alberta and Colorado in the lee of the Rocky Mountains, and in northern Argentina in the lee of the Andes. It also occurs off southeast Greenland where the barrier effect of the ice sheet promotes cyclogenesis in the Denmark Strait. The development of such lee cyclones contributes to the strength and position of the mean Icelandic low. 2 The thermal low These lows occur almost exclusively in summer, resulting from intense daytime heating of continental areas. Figure 7.1C illustrates their vertical structure. The most impressive examples are the summer low pressure cells over Saudi Arabia, the northern part of the Indian subcontinent, and Arizona. The Iberian Peninsula is another region commonly affected by such lows. They occur over southwestern Spain on 40–60 percent of days in July and August. Typically, their intensity is only 2–4mb and they extend to about 750mb, less than in other subtropical areas. The weather accompanying them is usually hot and dry, but if sufficient moisture is present the instability caused by heating may lead to showers and thunderstorms. Thermal lows normally disappear at night, when the heat source is cut off, but in fact those in India and Arizona persist. 3 Polar lows Polar lows are a loosely defined class of mesoscale to subsynoptic-scale systems (a few hundred kilometers across) with a lifetime of one to two days. On satellite imagery, they appear as a cloud spiral with one or several cloud bands, as a comma cloud (see Figure 9.17 and Plate 9.2), or as a swirl in cumulus cloud streets. They develop mainly in the winter months, when unstable mP or mA air currents stream equatorward along the

eastern side of a north–south ridge of high pressure, commonly in the rear of an occluding primary depression. They usually form within a baroclinic zone, e.g., near sea-ice margins, where there are strong sea surface temperature gradients, and their development may be stimulated by an initial upper-level disturbance. In the Northern Hemisphere, the comma cloud type (which is mainly a cold-core disturbance of the middle troposphere) is more common over the North Pacific, while the spiral-form polar low occurs more often in the Norwegian Sea. The latter is a low-level warmcore disturbance that may have a closed cyclonic circulation up to about 800mb or it may consist simply of one or more troughs embedded in the polar airflow. A key feature is the presence of an ascending, moist, southwesterly flow relative to the low center. This organization accentuates the general instability of the cold airstream to give considerable precipitation, often as snow. Latent heat release is an important mechanism for generating polar lows in the southern Norwegian Sea while stronger low-level baroclinicity and weaker convection prevail in systems in the northern Norwegian Sea and Barents Sea. Heat input to the cold air from the sea continues by

Plate 9.2 A polar low pressure system located in the Arctic Ocean on 25 February, 2008. On the right side of this

image are the Queen Elizabeth Islands, Canada. Polar lows are similar to tropical cyclones if they reach sufficient

strength but last only about 12 to 36 hours. They cause high surface winds and snow, and are often easily

recognizable in satellite images because of their characteristic pattern, sometimes called a ‘comma cloud’ because

of their hook-like shape.

Source: Jeff Schmaltz, Visible Earth, NASA. night and day, so in exposed coastal districts showers may occur at any time. In the Southern Hemisphere, polar low mesocyclones appear to be most frequent in the transition seasons, as these are the times of strongest meridional temperature and pressure gradients. In addition, over the Southern Ocean the patterns of occurrence and movement are more zonally distributed than in the Northern Hemisphere. 4 The cold low The cold low (or cold pool) is usually most evident in the circulation and temperature fields of the middle troposphere. Characteristically, it displays symmetrical

isotherms around the low center. Surface charts may show little or no sign of these persistent systems, which are frequent over the northeastern North America and northeast Siberia. They probably form as the result of strong vertical motion and adiabatic cooling in occluding baroclinic lows along the Arctic coastal margins. Such lows are especially important during the Arctic winter in that they bring large amounts of medium and high cloud, which offsets radiational cooling of the surface. Otherwise, they usually cause no ‘weather’ in the Arctic during this season. It is important to emphasize that tropospheric cold lows may be linked with either low or high pressure cells at the surface. In mid-latitudes, cold lows may form during periods of low-index circulation pattern (see Figure 6.27) by the cutting-off of polar air from the main body of cold air to the north (these are sometimes referred to as cut-off lows). This gives rise to weather of polar air-mass type, although rather weak fronts may also be present. Such lows are commonly slow-moving and give persistent unsettled weather with thunder in summer. Heavy 10 0 10 -2 10 -1 10 0 10 1 10 2 10 3 10 4 10 5 10 6 10 7 10 8 Minute Hour Day 2 days 4 days Week Month Year Jet stream Anticyclone Hurricane Local winds Thunderstorm Large cumulus Tornado Small cumulus Dust devil Distance scale (m) Molecular diffusion Sound waves L i f e s p a n (s e c) Small-scale turbulence 10 1 10 2 10 3 10 4 10 5 10 6 10 7 10 8 Figure 9.26 The spatial scale and lifespans of mesoscale and other meteorological systems.

precipitation over Colorado in spring and autumn

is often associated with cold lows.

I MESOSCALE CONVECTIVE

SYSTEMS

Mesoscale convective systems (MCSs) are

intermediate in size and lifespan between synoptic

disturbances and individual cumulonimbus

cells (see Figure 9.26). Figure 9.27 shows the

movement of clusters of convective cells, each cell

about 1km in diameter, as they crossed southern

Britain with a cold front. Each individual cell may

be short-lived, but cell clusters may persist for

hours, strengthening or weakening due to

orographic and other factors.

MCSs occur seasonally in mid-latitudes (parti

cularly the central United States, eastern China and

South Africa) and the tropics (India, West and

Central Africa and northern Australia) as either

nearly circular clusters of convective cells or linear

squall lines. The squall line consists of a narrow line

of thunderstorm cells, which may extend for

hundreds of kilometers. It is marked by a sharp

veer in wind direction and very gusty conditions. The squall line often occurs ahead of a cold front, maintained either as a self-propagating disturbance or by thunderstorm downdrafts. It may form a pseudo-cold front between rain-cooled air and a rainless zone within the same air mass. Midlatitude squall lines appear to form through one of two mechanisms: (1) a pressure jump that propagates as a bore; (2) the leading edge of a cold front aloft (CFA) acting on instability present to the east of an orographic lee trough. In frontal cyclones, cold air in the rear of the depression may overrun air in the warm sector. The intrusion of this nose of cold air sets up great instability, and the subsiding cold wedge tends to act as a scoop forcing up the slower moving warm air. Figure 9.28 demonstrates that the relative motion of the warm air is towards the squall line. Such conditions generate severe frontal thunderstorms such as that which struck Wokingham, England, in September 1959. This moved from the southwest at about 20m s −1 , steered by strong southwesterly flow aloft. The cold air subsided from high levels as a violent squall, and the updraft ahead of this produced an intense hailstorm. Hailstones grow by accretion in the upper part of the updraft, where speeds in excess of 50m s −1 are not uncommon, are blown ahead of the storm by strong upper winds, and begin to fall. This causes surface melting, but the stone is caught up again by the advancing squall line and reascends. The melted surface freezes, giving glazed

ice as the stone is carried above the freezing level, and further growth occurs through the collection of supercooled droplets (see also Chapter 4, pp. 124 and 140). Various types of MCS occur over the central United States in spring and summer (see Figure 9.29), bringing widespread severe weather. They may be small convective cells organized linearly, or as a large amorphous cell known as a mesoscale convective complex (MCC). This develops from initially isolated cumulonimbus cells. As rain falls from the thunderstorm clouds, evaporative cooling of the air beneath the cloud bases sets up cold downdrafts, and when these become 0 100 km 23.45 21.45 19.45 17.45

15.45

Figure 9.27 Successive positions of individual

clusters of mid-tropospheric convective cells

moving across southern Britain at about 50km hr −1

with a cold front. Cell location and intensity were

determined by radar.

Source: After Browning (1990). By permission of the Royal

Meteorological Society. sufficiently extensive they create a local high pressure of a few millibars' intensity. The downdrafts trigger the ascent of displaced warm air, and a general warming of the middle troposphere results from latent heat release. Inflow develops towards this warm region, above the cold outflow, causing additional convergence of moist, unstable air. In some cases, a low-level jet provides this inflow. As individual cells become organized in a cluster along the leading edge of the surface high, new cells tend to form on the right flank (in the Northern Hemisphere) through interaction of cold downdrafts with the surrounding air. Through this process and the decay of older cells on the left flank, the storm system tends to move 10–20° to the right of the mid-tropospheric wind direction. As the thunderstorm high intensifies, a 'wake low', associated with clearing weather, forms to the rear of it. The system is now producing violent winds, and intense downpours of rain and hail accompanied by thunder. During the triggering of new cells, tornadoes may form as discussed below. As the MCC reaches maturity during the evening and night hours over the Great Plains, the mesoscale circulation is capped by an extensive (>100,000km 2) cold upper-cloud shield, readily

identified on infrared satellite images. Statistics for 43 systems over the Great Plains in 1978 showed that the systems lasted on average 12 hours, with initial mesoscale organization occurring in the early evening (18:00–19:00 LST) and maximum extent seven hours later. During their life-cycle, systems may travel from the Colorado– Kansas border to the Mississippi River or the Great Lakes, or from the Missouri– Mississippi River valley to the east coast. An MCC usually decays when synoptic-scale features inhibit its self-propagation. The production of cold air is shut off when new convection ceases, weakening the meso-high and -low, and the rainfall becomes light and sporadic, eventually stopping altogether. Particularly severe thunderstorms are associated with great potential vertical instability (e.g., hot, moist air underlying dryer air, with colder air aloft). This was the case with a severe storm in the vicinity of Sydney, Australia, on 21 January 1991 (Figure 9.30). The storm formed in a hot, moist, low-level airstream flowing northeast on the eastern side of the Blue Mountains escarpment. This flow was overlain by a hot, dry, northerly airstream at an elevation of 1500–6000 meters, 0 0 5 10 15 10 20 30 40 50 5 5 8miles0 05 High altitude wind shear Cloud outline Moist warm updraughts Wind gust front Storm advancing Tornadoes form here Typical hail paths Squall line Hail Heavy rain Dry cold downdraughts 510 10 15km H e i g h t (, 0 0 0 s f e e t) H e i g h t (, 0 0 0 s m e t e r s) Figure 9.28 Thunder cell structure with hail and tornado formation. Source: After Hindley (1977). By permission of New Scientist.

which, in turn, was capped by cold air associated

with a nearby cold front. Five to seven such severe

thunderstorms occurred annually in the vicinity

of Sydney during 1950–1989.

On occasion, so-called supercell thunderstorms

may develop as new cells, forming downstream, are swept up by the movement of an older cell. These are about the same size as thunder cell clusters but are dominated by one giant updraft and localized strong downdrafts Figure 9.31). They may give rise to large hailstones and tornadoes, although some give only moderate Initiation LINEAR CONVECTIVE SYSTEMS OCCLUDING CONVECTIVE SYSTEMS CHAOTIC CONVECTIVE SYSTEMS Development Maturity

Figure 9.29 Schematic evolution of three convective modes on the US Great Plains showing several

scales of cloud development (shading).

Source: Blanchard (1990, p. 996, Fig.2). Courtesy American Meteorological Society. rainfall amounts. A useful measure of instability in mesoscale storms is the bulk Richardson Number (Ri) which is the (dimensionless) ratio of the suppression of turbulence by buoyancy to the generation of turbulence by vertical wind shear in the lower troposphere. A high value of Ri means weak shear compared to buoyancy; Ri > 45 favors independent cell formation away from the parent updraft. For Ri < 30, strong shear supports a supercell by keeping the updraft close to its downdraft. Ri < 10 indicates weak instability and for strong vertical shear. Tornadoes, which often develop within MCSs, are common over the Great Plains of the United States, especially in spring and early summer (see Figure 9.32 and Plate 9.3). During this period, cold, dry air from the high plateaux may override maritime tropical air (see Note 1). Subsidence beneath the upper tropospheric westerly jet (Figure 9.33) forms an inversion at about 1500–2000m, capping the low-level moist air. The moist air is extended northward by a low-level southerly jet (cf. p. 261) and, through continuing advection the air beneath the inversion becomes progressively more warm and moist. Eventually, the general convergence and ascent in the depression trigger the potential instability of the air, generating large cumulus clouds, which 0 100 km Newcastle Sydney HOT DRY 1500 – 6000m

Canberra 1 3 Tasman Sea 5 H O T M O I S T < 1 , 5 0 0 m

Figure 9.30 Conditions associated with the severe

thunderstorm near Sydney, Australia, on 21 January 1991.

The contours indicate the mean annual number of severe

thunderstorms (per 25,000km 2) over eastern New South

Wales for the period 1950–1989.

Source: Based on Griffiths et al. (1993). After Eyre (1992). Reproduced by

kind permission of the NSW Bureau of Meteorology, from Weather by

permission of the Royal Meteorological Society. Copyright ©. 15 10 Back-sheared anvil Mammatus Cold air Flaking line towers Precipitation-free cloud base Rear-flank gust front Precipitation curtain Warm air A i r f l o w Debris Tornado Large hail Small hail Heavy rain Light rain Horizon NE 0 50 km Forward-flank gust front Motion Wall cloud Overshooting top Mammatus Anvil Tail cloud 5 0 SW H e i g h t (k m) Figure 9.31 A supercell thunderstorm. Source: After the National Severe Storms Laboratory, USA and H. Bluestein; from Houze and Hobbs (1982). Copyright © Academic Press, reproduced by permission. 0.5 0.5 0.5 500 450 400 350 300 250 200 150 100 50 0 JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC 0 5 10 15 20 25 JAN FEB MAR M o n t h l y a v e r a g e U S T o r n a d o d e a t h s M o n t h l y a v e r a g e n u m b e r o f T o r n a d o e s APR MAY JUN JUL AUG SEP OCT NOV DEC 0.5 (A) (B) (C) 0.5 0.5 0.5 0.5 1 1 1 3 3 3 3 3 1 3 3 3 3 5 5 5 5 5 5 5 5 7 7 7 7 7 9 9 9

Figure 9.32 Tornado characteristics in the United States. A: Frequency of tornadoes (per 26,000km 2) in

the United States, 1953–1980. B: Monthly average number of tornadoes for each of the nine years shown

by the alternating black and white bars (1990–1998). C: Monthly averages of resulting deaths (1966–1995).

Sources: A: From NOAA (1982). B and C: After NOAA – Storm Prediction Center. Plate 9.3 A tornado raising surface dust in the Midwest USA. Source: Courtesy Mark Anderson, University of Nevada

penetrate the inversion. The convective trigger is

sometimes provided by the approach of a cold

front towards the western edge of the moist tongue. Tornadoes may also occur in association with tropical cyclones (see p. 341) and in other synoptic situations if the necessary vertical contrast is present in the temperature, moisture and wind fields. The exact tornado mechanism is still not fully understood owing to observational difficulties. Tornadoes tend to develop in the right-rear quadrant of a severe thunderstorm. Supercell thunderstorms are often identifiable in plan view on a radar reflectivity display by a hook echo pattern on the right-rear flank (Plate 9.4). The echo represents a (cyclonic or anticyclonic) spiral cloud band around a small central eye and its appearance may signal tornado development. The origin of the hook echo appears to involve

the R o c k y M o u n t a i n s Low-level Moist Tongue Low Gulf of Mexico H i g h l e v e l J e t L o w l e v e l J e t

Figure 9.33 The synoptic conditions favoring

severe storms and tornadoes over the Great Plains.

Plate 9.4 A radar image of a hook echo that is often a precursor of tornadic activity.

Source: NOAA NSS 0104. horizontal advection of precipitation from the rear of the mesocyclone. Rotation develops where a thunderstorm updraft interacts with the horizontal airflow. Provided that the wind speed increases with height, the vertical wind shear generates vorticity (Chapter 6C) around an axis normal to the airflow, which is then tilted vertically by the updraft. Directional shear also generates vorticity that the updraft translates vertically. These two elements lead to rotation in the updraft in the lower-middle troposphere forming a meso-low 10–20km across. Pressure in the meso-low is 2–5mb lower than in the surrounding environment. At low levels, horizontal convergence increases the vorticity and rising air is replenished by moist air from progressively lower levels as the vortex descends and intensifies. The meso-low shrinks in diameter and the conservation of momentum increases the wind speed. At some point, a tornado, sometimes with secondary vortices (Figure 9.34), forms within the meso-low. The tornado funnel has been observed to originate in the cloud base and extend towards the surface (Figure 9.34). One idea is that convergence beneath the base of cumulonimbus clouds, aided by the interaction between cold precipitation downdrafts and neighboring updrafts, may initiate the funnel. Other observations suggest that the funnel forms simultaneously throughout a considerable depth of cloud, usually a towering cumulus. The upper portion of the tornado spire in this cloud may become linked to the main updraft of a neighboring cumulonimbus, causing rapid removal of air from the spire and allowing a sharp pressure decrease at the surface. The pressure drop is estimated to exceed 200–250mb in some cases, and it is this that makes the funnel visible by causing air entering the vortex to reach saturation. Over water, tornadoes are termed waterspouts; the majority rarely attain extreme intensities. The tornado vortex is usually only a few hundred meters in diameter and in an even more restricted band around the core the winds can attain speeds of 50–100m s −1 . Intense tornadoes may have multiple vortices rotating anticlockwise with respect to

the main tornado axis, each following a cycloidal path. The whole tornado system gives a complex pattern of destruction, with maximum wind speeds on the right-side boundary (in the T T V V (Min. V-T : Max. V+T) Translational speed of tornado center Translational speed of suction vortex Rotational speed around tornado center Wind directions Tornado center Suction vortex L e f t s i d e d b o u n d a r y R i g h t s i d e d b o u n d a r y P a t h o f t o r n a d o c e n t e r V T V + T S u c t i o n s w a t h Figure 9.34 Schematic diagram of a complex tornado with multiple suction vortices. Source: After Fujita (1981, p.1251, fig. 15). Courtesy of the American Meteorological Society.

Plate 9.5 The super-tornado outbreak of 3–4 April 1974 when 148 tornadoes were reported in 24 hours. Thirty of

these tornadoes were classified as F4 or F5 (winds of 92–142ms −1) on the Fujita-Pearson Tornado Scale. There were

315 storm related fatalities, over 6100 storm-related injuries; and damages exceeded US$600 million (1974 dollars). Northern Hemisphere), where the translational and rotational speeds are combined. Destruction results not only from the high winds, because buildings near the path of the vortex may explode outward owing to the pressure reduction outside. Intense tornadoes present problems as to their energy supply, and it has recently been suggested that the release of heat energy by lightning and other electrical discharges may be an additional energy source. Tornadoes commonly occur in families and move along rather straight paths (typically between 10 and 100km long and 100m to 2km wide) at velocities dictated by the low-level jet. Thirty-year averages indicate some 750 tornadoes per year in the United States, with 60 percent of these during April to June (see Figure 9.32B). The largest outbreak in the United States occurred on 3–4 April 1974, extending from Alabama and Georgia in the south to Michigan in the north and from Illinois in the west to Virginia in the east. This ‘Super Outbreak’ spawned 148 tornadoes in 20 hours with a total path length of over 3200km (Plate 9.5). Tornadoes in the United States caused an average of 59 fatalities per year during 1975–2006 and about 1800 injuries annually, although most of the deaths and destruction result from a few long-lived tornadoes, making up only 1.5 percent of the total reported. For example, the most severe recorded tornado traveled 200km in three hours across Missouri, Illinois and Indiana on 18 March 1925, killing 689 people. Tornado fatalities were almost 19,000

during the period 1880–2005 but the annual rate has declined considerably from the 1920s-1930s. The highest proportion (44 percent) of fatalities occur in mobile homes. Tornadoes also occur in Canada, Europe, Australia, South Africa, India and East Asia. They are not unknown in the British Isles. During 1960–1982 there were 14 days per year with tornado occurrences. Most are minor outbreaks, but on 23 November 1981, 102 were reported during southwesterly flow ahead of a cold front. They are most common in autumn, when cold air moves over relatively warm seas.

Ideal air masses are defined in terms of barotropic conditions, where isobars and isotherms are

assumed to be parallel to each other and to the surface. The character of an air mass is determined

by the nature of the source area, changes due to air-mass movement and its age. On a regional

scale, energy exchanges and vertical mixing lead to a measure of equilibrium between surface

conditions and those of the overlying air, particularly in quasi-stationary high pressure systems.

Air masses are conventionally identified in terms of temperature characteristics (arctic, polar,

tropical) and source region (maritime, continental). Primary air masses originate in regions of semi

permanent anticyclonic subsidence over extensive surfaces of similiar properties. Cold air masses

originate either in winter continental anticyclones (Siberia and Canada), where snow cover

promotes low temperatures and stable stratification, or over high-latitude sea ice. Some sources

are seasonal, like Siberia; others are permanent, such as Antarctica. Warm air masses originate

either in shallow tropical continental sources in summer or as deep, moist layers over tropical

oceans. Air-mass movement causes stability changes through thermodynamic processes

(heating/cooling from below and moisture exchanges) and by dynamic processes (mixing,

lifting/subsidence), producing secondary air masses (e.g., mP air). The age of an air mass determines

the degree to which it has lost its identity as the result of mixing with other air masses and vertical

exchanges with the underlying surface. Air-mass boundaries give rise to baroclinic frontal zones a few hundred kilometers wide. The

classical (Norwegian) theory of mid-latitude cyclones considers that fronts are a key feature of their

formation and life-cycle. Newer models show that instead of the frontal occlusion process, the warm

front may become bent back with warm air seclusion within the polar airstream. Cyclones tend to

form along major frontal zones – the polar fronts of the North Atlantic and North Pacific regions

and of the southern oceans. An Arctic front lies poleward and there is a winter frontal zone over

the Mediterranean. Air masses and frontal zones move poleward (equatorward) in summer (winter). Newer cyclone theories regard fronts as rather incidental. Cloud bands and precipitation areas

are associated primarily with conveyor belts of warm air. Divergence of air in the upper troposphere

is essential for large-scale uplift and low-level convergence. Surface cyclogenesis is therefore

favored on the eastern limb of an upper wave trough. ‘Explosive’ cyclogenesis appears to be

associated with strong wintertime gradients of sea surface temperature. Cyclones are basically

steered by the quasi-stationary long (Rossby) waves in the hemispheric westerlies, the positions

of which are strongly influenced by surface features (major

mountain barriers and land–sea surface

temperature contrasts). Upper baroclinic zones are associated with jet streams at 300–200mb, which

also follow the longwave pattern. The idealized weather sequence in an eastward-moving frontal depression involves increasing

cloudiness and precipitation with an approaching warm front; the degree of activity depends on

whether the warm-sector air is rising or not (ana- or kata-fronts, respectively). The following cold

front is often marked by a narrow band of convective precipitation, but rain both ahead of the warm

front and in the warm sector may also be organized into locally intense mesoscale cells and bands

due to the ‘conveyor belt’ of air in the warm sector. Some low pressure systems form through non-frontal mechanisms. These include the lee

cyclones formed in the lee of mountain ranges; thermal lows due to summer heating; polar air

depressions commonly formed in an outbreak of maritime arctic air over oceans; and the upper

cold low, which is often a cut-off system in upper wave development or an occluded mid-latitude

cyclone in the Arctic. Mesoscale convective systems (MCSs) have a spatial scale of tens of kilometers and a timescale

of a few hours. They may give rise to severe weather, including thunderstorms and tornadoes.

Thunderstorms are generated by convective uplift, which may result from daytime heating,

orographic ascent or squall lines. Several cells may be organized in a mesoscale convective

complex and move with the large-scale flow. Thunderstorms associated with a moving convective

system provide an environment for hailstone growth and for

the generation of tornadoes.

• What are the essential differences between mesoscale and synoptic scale systems?

• Using an appropriate website with synoptic weather maps (see Appendix 4D), trace the movement of frontal and non-frontal lows/troughs and high pressure cells over a five-day period, determining rates of displacement and changes of intensity of the systems.

• In the same manner, examine the relationship of surface lows and highs to features at the 500mb level.

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10 Weather and climate in middle and high latitudes

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11 Tropical weather and climate

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• Consider the various factors that influence the damage caused by a tropical cyclone upon landfall in different parts of the world (e.g., the southeastern USA, islands in the Caribbean, Bangladesh, Northern Australia and Hong Kong).

• Use the indices of ENSO, NAO, PNA and so on available on the Web (see Appendix 4D) to compare anomalies of temperature and precipitation in a region of interest to you during positive and negative phases of the oscillations).

• Examine the similarities and differences of the major monsoon climates of the world.

• What are the similarities and differences of cyclonic systems in mid-latitudes and the tropics?

• By what mechanisms do ENSO events affect weather anomalies in the tropics and in other parts of the world?

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smog and inhibits both incoming and outgoing radiation. Pollution domes and plumes are produced

around cities under appropriate conditions of vertical temperature structure and wind velocity. H

and G dominate the urban heat budget, except in city parks, and as much as 70–80 percent of

incoming radiation may become sensible heat, which is very variably distributed between the

complex urban built forms. Urban influences combine to give generally higher temperatures than

in the surrounding countryside, not least due to the growing importance of heat production by

human activities. These factors lead to the urban heat island, which may be 6–8°C warmer than

surrounding areas in the early hours of calm, clear nights, when heat stored by urban surfaces is

being released. The urban–rural temperature difference under calm conditions is statistically related

to the city population size; the urban canyon geometry and sky view are major controlling factors.

The heat island may be a few hundred meters deep, depending on the building configuration. In

some cases, summer daytime cold islands are observed. Urban wind speeds are generally lower

than in rural areas by day, but the wind flow is complex, depending on the geometry of city

structures. Cities tend to be less humid than rural areas, but their topography, roughness and

thermal qualities can intensify summer convective activity over and downwind of the urban area,

giving more thunderstorms and heavier storm rainfall. Tropical cities have heat islands, but the

diurnal phase tends to be delayed relative to mid-latitude ones. The temperature amplitude is largest

during dry season conditions.

• In what ways do vegetated surfaces modify the surface climate compared with unvegetated ones and what processes are involved?

• What are the major effects of urban environments on atmospheric composition? (Data from air sampling sites in North America and Europe are available on the Web.)

• Look for evidence from local weather station reports and/or vegetation types for topoclimatic differences in locations where you live/travel and consider whether these arise from differences in solar radiation, day/night temperatures, moisture balance, wind speed or combinations of these factors.

• Look for evidence of urban–rural climatic differences in cities near you using weather reports of day/night temperatures, visibility, snowfall events and so on.

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13 Climate change

• Examine the figures showing climatic time series in Chapter 13 for evidence of changes in mean and variance and consider where step function changes have occurred and where trends can be detected.

• What are the different climate forcing factors at work on geological and historical timescales?

• What are the main advantages and limitations of different types of proxy records of paleoclimate? Consider the climatic variables that may be inferred and the temporal resolution of the information.

• What are the main reasons for uncertainties in projections of climate for the year 2100?

• What are some of the possible impacts of projected climate changes in your region and country? The magnitude of the response of global temperature to a radiative forcing of given magnitude,

or set of forcings in combination, depends on the climate feedbacks. Positive feedbacks dominate,

and hence act to amplify the temperature response to a forcing. In terms of human-induced climate

change, the most important are the fast water vapor and ice-albedo feedbacks. Climate projections through the twenty-first century, assuming a variety of emission scenarios

for greenhouse gases and aerosols, indicate a mean global temperature increase in the range

of 2–4 o C by the year 2100, together with sea-level rises of 200–500mm. Given the rapid growth of

greenhouse gas concentrations in recent decades, the effects of ice sheet dynamics and other wild

cards in the system, these may be underestimates. The Arctic will eventually become free of sea

ice in summer. Warming will also be accompanied by continued shrinking of glaciers, ice caps and

permafrost, changes in the hydrologic cycle, atmospheric

circulation, and vegetation. Critical research needs include improved understanding of climate feedbacks, including

feedbacks in the carbon cycle, and the role of the oceans in the uptake of heat and carbon dioxide.

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APPENDIX 1 Climate classification

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APPENDIX 4 Data sources

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